

Human receptive endometrial assembloid for deciphering the implantation window

Reviewed Preprint

v3 • September 19, 2025

Revised by authors

Reviewed Preprint

v2 • July 19, 2024

Reviewed Preprint

v1 • October 30, 2023

Yu Zhang, Rusong Zhao, Chaoyan Yang, Jinzhu Song, Peishu Liu, Yan Li, Boyang Liu, Tao Li, Changjian Yin, Minghui Lu, Zhenzhen Hou, Chuanxin Zhang, Zi-Jiang Chen , Keliang Wu , Han Zhao 

State Key Laboratory of Reproductive Medicine and Offspring Health, Center for Reproductive Medicine, Institute of Women, Children and Reproductive Health, Cheeloo College of Medicine, Shandong University, Nanjing, China • National Research Center for Assisted Reproductive Technology and Reproductive Genetics, Shandong University, Jinan, China • Key Laboratory of Reproductive Endocrinology (Shandong University), Ministry of Education, Jinan, China • Shandong Technology Innovation Center for Reproductive Health, Jinan, China • Shandong Provincial Clinical Research Center for Reproductive Health, Jinan, China • The Affiliated Suzhou Hospital of Nanjing Medical University, Suzhou Municipal Hospital, Gusu School, Nanjing Medical University, Suzhou, China • Shandong Key Laboratory of Reproductive Medicine, Shandong Provincial Hospital Affiliated to Shandong First Medical University, Jinan, China • Research Unit of Gametogenesis and Health of ART-Offspring, Chinese Academy of Medical Sciences (No.2021RU001), Jinan, China • Department of Obstetrics and Gynecology, Qilu Hospital of Shandong University, Jinan, China • Department of Gynecology, Shandong Provincial Hospital, Cheeloo College of Medicine, Shandong University, Jinan, China • Department of Gynecology, Shandong Provincial Hospital Affiliated to Shandong First Medical University, Jinan, China • Shanghai Key Laboratory for Assisted Reproduction and Reproductive Genetics, Shanghai, China • Department of Reproductive Medicine, Ren Ji Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai, China

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This **important** study reports an endometrial organoid culture system mimicking the window of implantation. The evidence supporting the conclusion drawn is **convincing**. The data will be of interest to embryologists and investigators working on reproductive biology and medicine.

<https://doi.org/10.7554/eLife.90729.3.sa3>

Abstract

Human endometrial receptivity is a critical determinant of pregnancy success; however, in vivo studies of its features and regulation are particularly challenging due to ethical restriction. Recently, the development of human endometrial assembloids has provided a powerful model to examine this intricate biological process. In this study, we established a specialized human window-of-implantation (WOI) endometrial assembloid system that mimics the in vivo receptive endometrium. It recapitulates not only the structural attributes of pinopodes and cilia, but also molecular characteristics of mid secretory endometrium. Furthermore, the WOI endometrial assembloid exhibits hormone responsiveness, energy

metabolism with larger and enhanced functional mitochondria, increased ciliary assembly and motility, and epithelial-mesenchymal transition (EMT), as well as promising potential for embryo implantation. As such, WOI assembloids hold great promise as a platform to unravel the intricate mechanisms governing endometrial receptivity regulation, maternal-fetal interactions, and associated diseases, ultimately driving impactful advancements in the field.

Introduction

The human endometrium, comprising various cell types, experiences shedding (menstrual phase), regeneration (proliferative phase), and differentiation (secretory phase) under the regulation of estrogen and progesterone during the menstrual cycle (1). A brief window, called the “window-of-implantation (WOI)” or mid-secretory phase, allows for embryo implantation into the endometrium (1). The endometrium at the embryo implantation site contains diverse cells, including ciliated epithelial cells, decidualized stromal cells and immune cells. Cyclic endometrial changes akin to those in humans are exclusive to apes, Old World monkeys, molossid bats, and spiny mice, but not mice (1), rendering typical mouse models insufficient for accurately simulating human endometrium. Existing endometrial cell lines, such as Ishikawa and HEEC cells, consist solely of a single cell type, thus failing to reproduce the intricate physiological structure and function of the endometrium.

Replicating and reconstructing human organs has become essential for exploring tissue physiology and function. Organoid, a self-assembled 3D structure, closely resemble in vivo tissue or organ (2). They offer high expansibility, phenotypic, and functional properties, emerging as powerful tools for investigating tissue physiology and disease (3). In 2017, the first long-term and hormone-responsive human endometrial organoid was established using adult stem cells from endometrial biopsy (2, 3). Based on this, Margherita Y. Turco, et al developed noninvasive methods to construct endometrial organoids from menstrual flow (4). Takahiro Arima et al. developed polar reversal endometrial organoids to study embryo implantation (5). Apart from adult stem cells, pluripotent stem cells were also induced to endometrial stromal fibroblasts and epithelium, and then cocultured to form organoids (6, 7), which offered vigorous proliferative capacity but lacked immune cells and other components of the microenvironment. There are also studies that add immune cells to endometrial organoids for co-culture (8). In efforts to mimic human endometrial cell types in vitro, the endometrial organoids progressed from epithelial organoids (9), to assemblies of epithelial and stromal cells (10, 11) and then to stem cell-laden 3D artificial endometrium (12, 13), which were solely closer but not completely identical to the endometrium. Additionally, pathological endometrial organoid models have been established for conditions like endometriosis, endometrial hyperplasia, Lynch syndrome, and endometrial cancer (14). These organoids simulated endometrial morphology, hormone responsiveness, and physiological and pathological processes in vitro (3, 9, 15–17), facilitating the study of physiological phenomena (15, 18, 19), pathogenic mechanisms (16) and drug screening (14).

Although various regulators have been suggested to be involved in WOI, we are far from understanding how embryo implantation occurs during the WOI due to ethical limitations and fewer in vitro receptive endometrial model. Transforming from proliferative to receptive endometrium involves dynamic changes like decidualization, epithelial-mesenchymal transition (EMT) and ciliated epithelial development (1, 20), which have not been reconstructed in most in vitro model to date. Here, we successfully established a receptive endometrial assembloid system regulated by hormones and reflecting the in vivo WOI endometrial features, providing a platform for studying physiological and pathological endometrium and maternal-fetal interactions.

Results

Developing receptive endometrial assembloids in vitro

To establish endometrial assembloids in vitro, pre-receptive endometrium from reproductive-age women was dissociated into single cells or small cell masses. These cells then self-assembled into assembloids induced by various small molecules, such as Noggin, EGF, FGF2, WNT-3A and R-Spondin1 in expansion medium (ExM) (**Fig. 1A~B**, **Fig. S1A**). The assembloids derived from the first generation are used for experiments (**Fig. S1B**). The endometrial assembloids consist of vesicle-like glands and fibrous stromal cells (**Fig. S1C**, Video S1~S4). The epithelium marker E-cadherin, stromal cell marker vimentin, and endometrial gland marker FOXA2 were expressed in the cultured endometrial assembloid, which resembled the endometrium morphologically (**Fig. S1D**). Moreover, the endometrial assembloids exhibited substantial expression of the proliferation marker Ki67, while the apoptosis marker cleaved caspase-3 was undetectable, signifying the assembloids' robust proliferative capacity (**Fig. S1E**).

Functionally, endometrial assembloids effectively secreted glycogen into the lumen, mirroring the in vivo endometrial activity, thereby providing nourishment for embryo implantation (**Fig. S1D**). Moreover, after two days of estrogen (E2) treatment followed by fourteen days of medroxyprogesterone acetate (MPA) and cAMP administration, the assembloids exhibited a significant upregulation of progesterone receptor (PRA/B) expression, a modest increase in estrogen receptor α (ER α), and enhanced expression of estrogen-responsive genes *EGR1* and *OLFM4*, along with progesterone-responsive genes *PGR* and *PAEP* at the transcriptional level (**Fig. S1F~S1G**). It reflected a comparable hormonal responsiveness in the in vivo endometrium.

To identify hormonal regimens that induce implantation window, pregnancy-related hormones were supplemented into the culture system following the two-day E2. E2 and MPA promote the transition of endometrial assembloids into the secretory phase, while pregnancy hormones can promote the further differentiation. Prolactin (PRL) promotes immune regulation and angiogenesis during implantation (2, 21). Human chorionic gonadotropin (hCG) improves endometrial thickness and receptivity (22, 23). Human placental lactogen (hPL) promotes the development and function of endometrial glands (24). Hormone dosage was primarily based on peri-pregnant maternal body or localized endometrium levels (2). A comparison of multiple groups revealed similar counts, area, and average intensity of assembloids over time (**Fig. 1C-D**, **Fig. S1H**). However, only the final recipe (i.e., a combination of E2, MPA, cAMP, PRL, hCG and hPL) exhibited an endometrial receptivity-related gene expression profile, which highly expressed genes positively correlated with endometrial receptivity, and lowly expressed genes negatively correlated with receptivity, compared to the other hormone formulations (**Fig. 1E**). The assembloids induced with this scheme were defined as WOI assembloids (**Fig. 1A**). In contrast, endometrial assembloids maintained in ExM constituted the “control (CTRL)” group, whereas assembloids treated with E2 for two days, followed by E2, MPA, and cAMP for an additional six days, were induced to the secretory phase as previously reported (9), and designated as the “secretory (SEC)” group (**Fig. 1A**). There was no significant difference in the morphology of assembloids among the three groups (**Fig. 1B**), but when comparing receptivity markers, WOI assembloids exhibited high expression of IGFBP1, MAOA, and DPP4 (**Fig. 1F**), with increased glycogen secretion (**Fig. S1I**). Theoretically, the WOI assembloids are initially in the secretory phase, thus sharing characteristics with the SEC assembloids. More importantly, they are expected to display traits typical of the mid-secretory phase.

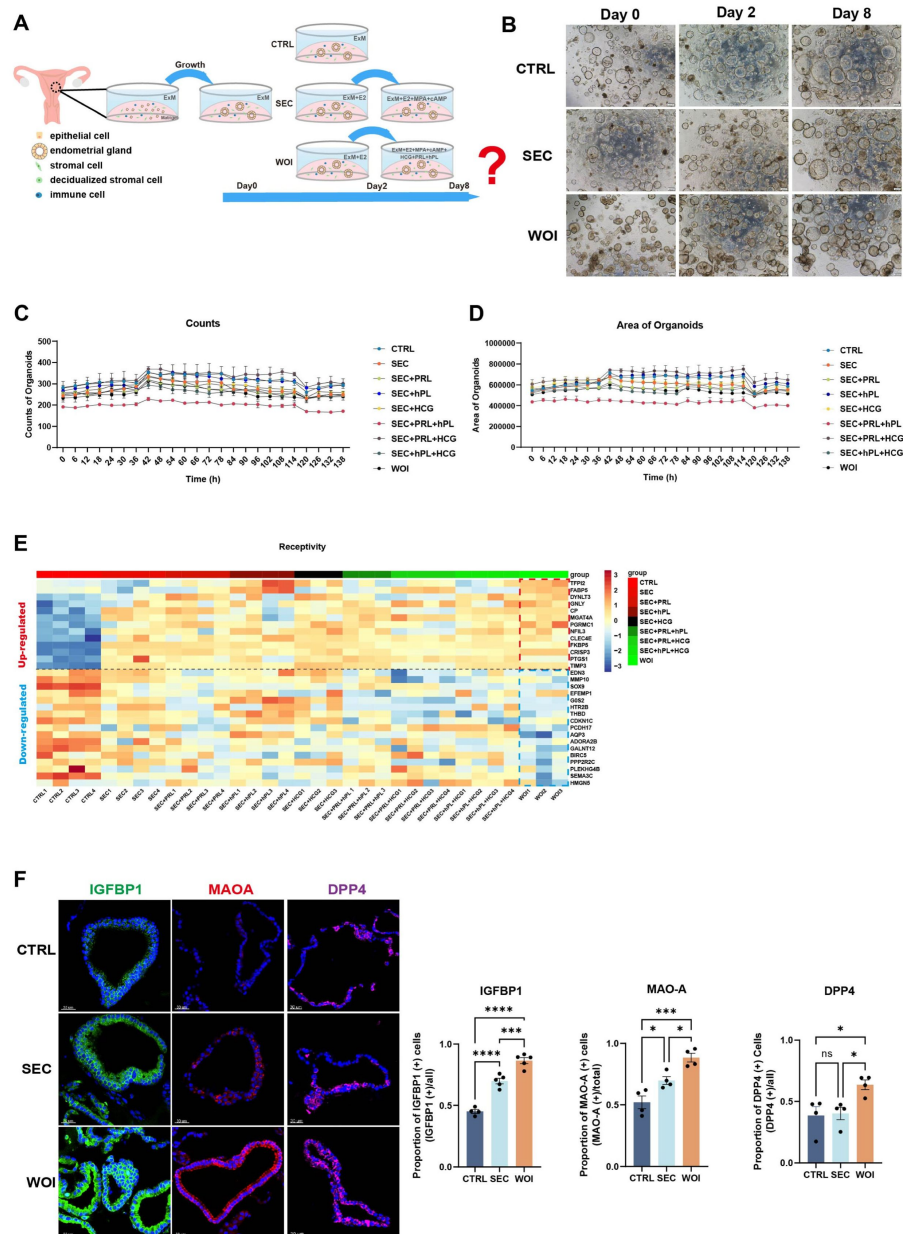


Fig 1

Developing receptive endometrial assembloids in vitro

(A) Human endometrial assembloids constructed from adult stem cells were treated with expansion medium (ExM) (CTRL) or subjected to hormonal stimulation. Timeline of endometrial assembloid cultured by ExM (CTRL), ovarian steroid hormones simulating secretory phase (SEC), ovarian steroid hormones combining PRL and placental hormones to mimic the window of implantation (WOI).

(B) Endometrial assembloids in the CTRL, SEC and WOI groups displayed similar growth patterns during the culture period. Scale bar = 200 μ m.

(C) The dynamic changes of the counts of assembloids over time in each hormone regimen.

(D) The dynamic changes of the area of assembloids over time in each hormone regimen.

(E) Heatmap showing receptivity related gene expression profile of assembloids in each hormone regimen. The color represents log-transformed fold change of gene expression.

(F) Validation of receptivity markers (IGFBP1, MAOA and DPP4) with immunofluorescence (IF) in the CTRL, SEC and WOI endometrial assembloids in vitro. Nuclei were counterstained with DAPI. Scale bar = 30 μ m. The bar chart displaying the quantitative comparison of receptivity markers among three groups. * $P \leq 0.05$, ** $P \leq 0.005$, *** $P \leq 0.0005$, **** $P \leq 0.0001$.

Receptive endometrial assembloids mimicked the implantation - window endometrium

Single-cell transcriptomics analysis identified the presence of epithelium, stromal cells, and immune cells in WOI assembloids with reference to CellMarker, PanglaoDB, Human Cell Atlas, Human Cell Landscape, and scRNASeqDB, and previous endometrium related studies(1, 9, 11, 25) (Fig. 2A, Fig. S2A~S2B). The WOI assembloids exhibited similarities to the mid-secretory endometrium in terms of glandular and luminal epithelium, secretory epithelium, LGR5 epithelium, ciliated epithelium, and EMT-derived cells, as evidenced by comparisons of scRNA-seq data from our assembloids and the mid-secretory endometrium as described by Stephen R. Quake in 2020 (1) (Fig. 2A, Fig. S2C~S2D). The morphology of immune and stromal cells was analyzed through 3D clearing staining and light sheet microscopy imaging, with vimentin labeling stromal cells (Vimentin⁺ or Vimentin⁺ F-actin⁺), CD45 and CD44 indicating immune cells, and FOXA2 identifying glands (Fig. 2B~2D). Furtherly, we confirmed the presence and types of immune cells using flow cytometry. White blood cells (WBC) were identified as CD45⁺ cells, with T cells, macrophages and NK cells characterized as CD45⁺CD3⁺ cells, CD45⁺CD68⁺CD11b⁺ cells and CD56⁺CD16⁻ cells, respectively (Fig. 2E).

The WOI assembloids displayed characteristic features of the receptive endometrium. On the one hand, the WOI assembloids secreted more glycogen into the lumen (Fig. S1I). On the other hand, the WOI assembloids possessed various characteristic microstructure of implantation window, including elongated microvilli and increased glycogen, pinopodes, and especially cilia (Fig. 2F).

Further, we concentrated on the comparison of gene expression and transcriptional regulation during the mid-secretory phase, as this phase marks the implantation window and involves substantial transcriptional alterations (Fig. 2G~2H). We referred to the scRNA-seq data of the mid-secretory phase from Stephen R. Quake 2020 (1). Pathways related to mitochondrial energy metabolism and cell adhesion exhibited upregulation in both the WOI assembloid and mid-secretory endometrium in comparison to the CTRL and SEC assembloids (Fig. 2G). The crucial transcription factors (TFs) found in the secretory epithelium and EMT-derived cells, which are implicated in implantation, revealed similarities between WOI assembloids and mid-secretory endometrium. Specifically, the secretory epithelium exhibited comparable TFs related to hypoxia response (such as previously reported HIF1A(26)), embryo implantation (such as FBLN1(27)), lipid metabolism (such as VMP1(28)), cell migration, and cell junction (such as TJP1(29)) (Fig. 2H). Similarly, EMT-derived cells also expressed similar TFs involved in endometrial decidualization (such as S100A10(30)), EMT (such as FAT1(31) and FOXF2(32)), and receptivity (such as NEAT1(33), SERPINB9(34), SOX17(35) and SOX4(36)) (Fig. 2H).

We then conducted the endometrial receptivity test (ERT) (37) to further assess the receptive state of WOI assembloids. ERT is a kind of gene analysis-based method for detecting endometrial receptivity, which combines high-throughput sequencing and machine learning to analyze the expression of endometrial receptivity-related genes (37). It is currently used in clinical practice to determine endometrial receptivity and guide personalized embryo transfer. The WOI assembloids derived from pre-receptive endometrium were observed to transit into the receptivity phase (Fig. S1J).

Collectively, these results demonstrated that the WOI assembloids closely resemble the in vivo endometrium during the implantation window in terms of structures and molecular characteristics.

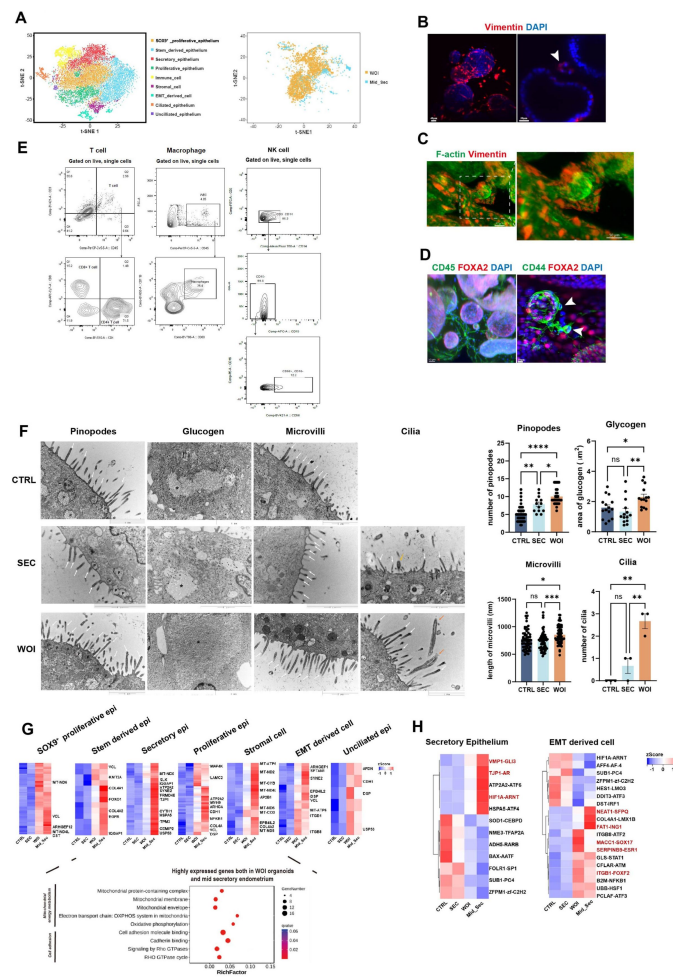


Fig 2

Receptive endometrial assembloids mimicked the implantation - window endometrium

(A) T-SNE plot of scRNA-seq data from three individual endometrial assembloids of the CTRL, SEC and WOI groups (left). T-SNE plot of combined scRNA-seq data from the three kinds of assembloids and mid-secretory endometrium (right).

(B) Exhibition of stromal cell marked by vimentin of CTRL assembloid through whole-mount clearing, immunostaining and light sheet microscopy imaging. Nuclei were counterstained with DAPI. The arrowhead indicates stromal cells. Scale bar = 40 μ m (left), Scale bar = 30 μ m (right).

(C) Whole-mount immunofluorescence showed that Vimentin⁺ F-actin⁺ cells (stromal cells) were arranged around the glandular spheres that were only F-actin⁺. Scale bar = 50 μ m.

(D) Exhibition of immune cell marked by CD45 and CD44, and endometrial gland marked by FOXA2 of CTRL assembloid through whole-mount clearing, immunostaining and light sheet microscopy imaging. Nuclei were counterstained with DAPI. The arrowhead indicates immune cells. Scale bar = 50 μ m (left), Scale bar = 10 μ m (right).

(E) Flow cytometric analysis of T cells and macrophages in the CTRL endometrial assembloid. Gating strategy used for determining white blood cells (WBC) (CD45⁺ cells), T cells (CD45⁺CD3⁺ cells) and macrophages (CD45⁺CD68⁺CD11b⁺ cells).

(F) Electron micrograph of the CTRL (top), SEC (middle) and WOI (bottom) endometrial assembloid showing pinopodes (P), glycogen granule (asterisk), microvilli (white arrows) and cilia (orange arrows). Scale bar = 1 μ m. Quantitative comparison of pinopodes, glycogen, microvilli, and cilia in the CTRL, SEC and WOI assembloids. *P $\leq$ 0.05, **P $\leq$ 0.005, ***P $\leq$ 0.0005, ****P $\leq$ 0.0001.

(G) Heatmap and bubble diagram illustrating highly expressed genes as well as GO functions enriched in both assembloids during the WOI and mid secretory endometrial tissue in terms of SOX9⁺ proliferative epithelium, stem-derived epithelium, secretory epithelium, proliferative epithelium, unciliated epithelium, stromal cells and EMT-derived cells. The color of heatmap represents log-transformed fold change of gene expression.

(H) Heatmaps showing differentially expressed TFs of endometrial assembloids and endometrium in the secretory epithelium (left) and EMT-derived cells (right). The color represents log-transformed fold change of gene expression.

Receptive endometrial assembloids recapitulate WOI-associated hormone response

We analyzed the transcriptome and proteome profiles of WOI assembloids compared to those of CTRL or SEC assembloids to assess WOI associated biological characteristics (**Fig. 3A** [↗](#), **Fig. S3A** [↗](#)~**S3E** [↗](#)).

WOI assembloids exhibited a robust hormone response, as demonstrated by the upregulation of PGR at the transcriptome level (**Fig. 3B** [↗](#)). Integrated analysis of the transcriptome and proteome data identified 179 upregulated genes/proteins in WOI assembloids compared to the CTRL group, with most of these genes/proteins implicated in the estrogen response (**Fig. S3F** [↗](#)~**S3G** [↗](#)). Furthermore, the analysis of progesterone response levels using PRA/B immunostaining indicated the highest levels in the WOI group, accompanied by an upregulation of the estrogen responsive protein OLFM4 (**Fig. 3C** [↗](#)). FOXO1, a crucial marker of endometrial receptivity reliant on PGR signaling, exhibited significantly elevated expression in WOI assembloids compared to CTRL assembloids (**Fig. 3D** [↗](#), Video S5), suggesting the involvement of progesterone signaling in the establishment of WOI assembloids. These findings collectively demonstrated a strong response of WOI assembloids to estrogen and progesterone.

Hormone response increased secretory epithelium and decreased proliferative epithelium in the SEC and WOI assembloids, suggesting the transformation from proliferation to secretory phase (**Fig. 3E** [↗](#)). The secretory epithelium, critical for the implantation window, contributed to cellular metabolic processes and HIF-1 signaling pathway response to hypoxia at the single-cell level (**Fig. S2E** [↗](#)~**S2F** [↗](#)). The secretory epithelium of WOI assembloids exhibited enhanced peptide metabolism and mitochondrial energy metabolism compared to the SEC group, supporting endometrial decidualization and embryo implantation (**Fig. 3F** [↗](#)). Proliferative epithelium differentiates into secretory epithelium under the regulation of branch nodal genes between state 5 and state 6, such as the KRT19, MALAT1, MT2A, RPL and RPS families, as revealed by single-cell trajectory (**Fig. 3G** [↗](#), **Fig. S2H** [↗](#)~**S2J** [↗](#)). The WOI assembloids showed more thoroughly differentiation from proliferative to secretory epithelium in relation to the SEC assembloids (**Fig. 3G** [↗](#)).

Overall, the WOI assembloids possessed the hormone response characteristic of implantation window, closely resembling the gene traits of embryo implantation.

Receptive endometrial assembloids possess enhanced energy metabolism

The WOI assembloids exhibited upregulation of monocarboxylic acid and lipid metabolism (represented by SLC25A1(**38** [↗](#))), and hypoxia response (represented by HIF1α(**26** [↗](#))) (**Fig. 3B** [↗](#), **Fig. S3G** [↗](#)~**S3I** [↗](#)). Likewise, the secretory epithelium, critical for the implantation window, contributed to cellular metabolic processes and HIF-1 signaling pathway response to hypoxia at the single-cell level (**Fig. S2E** [↗](#)~**S2F** [↗](#)).

To further investigate this trend, the Mfuzz algorithm was utilized to analyze gene expression across these three groups, focusing on gene clusters that were progressively upregulated or downregulated. It was observed that mitochondrial genes exhibited the highest expression levels in WOI endometrial assembloids (**Fig. 4A** [↗](#)). Additionally, at the protein level, WOI endometrial assembloids showed sustained high expression of mitochondrial proteins compared to SEC assembloids (**Fig. 4B** [↗](#)). TEM analysis revealed that WOI endometrial assembloids had the largest average mitochondrial area (**Fig. 4C** [↗](#)). The expression of mitochondrial-related genes increased from CTRL to SEC to WOI assembloids, with COA1 ensuring proper nuclear-mitochondrial connection(**39** [↗](#)), OXA1L promoting mitochondrial translation(**40** [↗](#)), and TIMMDC1 being crucial

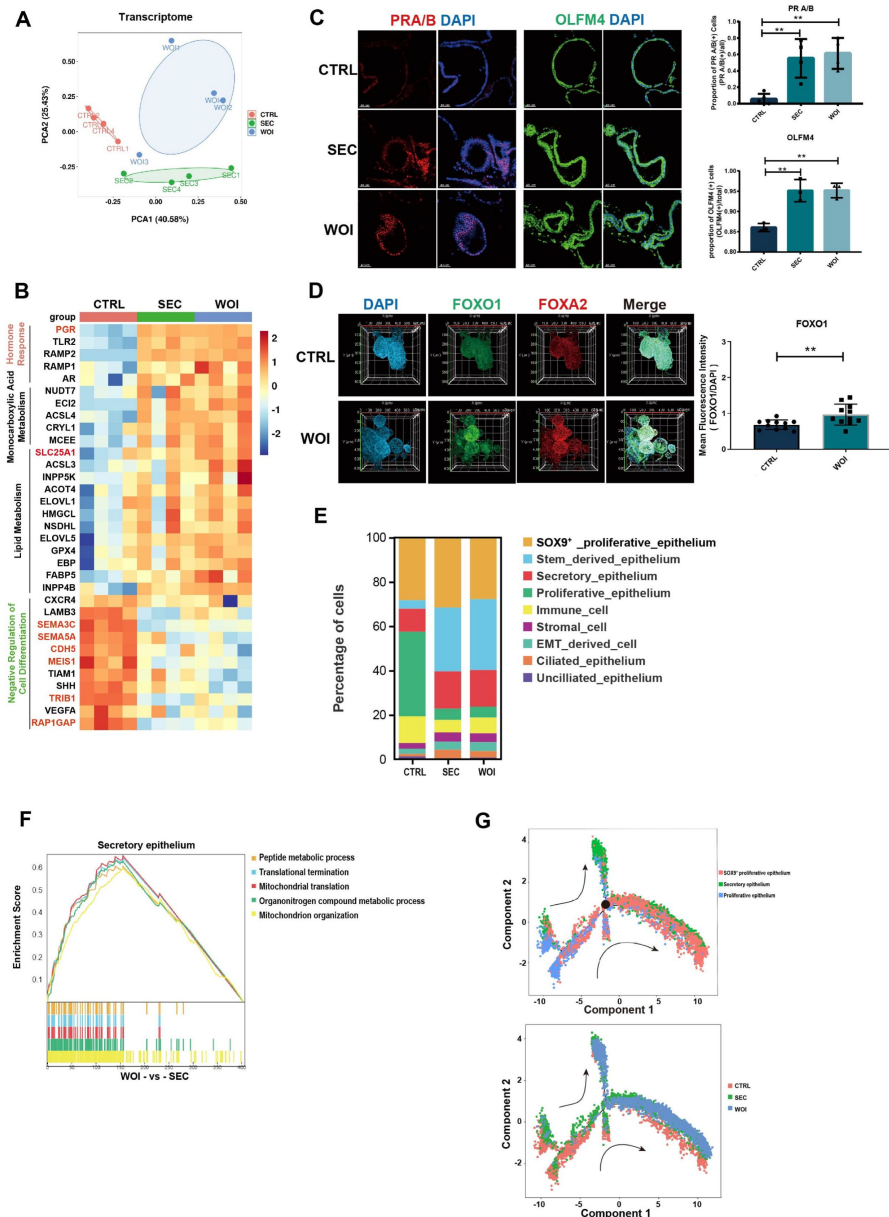


Fig 3

Receptive endometrial assembloids recapitulate WOI-associated hormone response

- (A) Principal component analysis (PCA) plot computed with differentially expressed genes in the bulk transcriptome of endometrial assembloids belonging to the CTRL, SEC and WOI groups.
- (B) Heatmap showing that enrichment of differentially expressed genes for the terms of hormone response, monocarboxylic acid metabolism, lipid metabolism, and negative regulation of cell differentiation. The color represents log-transformed fold change of gene expression.
- (C) Responsiveness to progesterone and estrogen was evaluated by IF to PRA/B and OLFM4 with IF respectively. Scale bar = 40 μ m, $^{**}P \leq 0.005$.
- (D) Exhibition of implantation marker (FOXO1) and endometrial gland marker (FOXA2) through combination of assembloid clearing, IF and light sheet microscopy. Nuclei were counterstained with DAPI. $^{**}P \leq 0.005$.
- (E) Bar graph exhibiting various percentages of each cell type in the three groups.
- (F) GSEA between the SEC and WOI groups for secretory epithelium.
- (G) Pseudotime trajectory showing the transformation between proliferative and secretory epithelium in the CTRL, SEC and WOI groups. Arrows indicate the direction of the pseudotime trajectory. The black dot indicates the key differentiation node.

for mitochondrial complex I assembly(41) (Fig. 4D). The WOI assembloids notably expressed higher levels of OXA1L and TIMMDC1 than the SEC assembloids (Fig. 4D). Furthermore, WOI assembloids produced more ATP and IL8(42) (Fig. 4E).

Thus, compared to SEC assembloids, WOI assembloids exhibited increased energy metabolism with larger and enhanced functional mitochondria.

Receptive endometrial assembloids increased the ciliary assembly and motility

The growth and development, assembly and movement of cilia, a characteristic endometrial structure, are essential for the establishment of endometrial implantation window and embryo implantation. Under the electron microscopy, the cilia were most observed in the WOI assembloids (Fig. 2E). Cilia-related genes and proteins exhibited the highest expression levels in WOI endometrial assembloids (Fig. 5A~5B). Compared to SEC assembloids, WOI endometrial assembloids showed upregulation in ciliary assembly, ciliary basal body, and motile cilia-related gene expressions, while genes related to non-motile cilia were downregulated (Fig. 5A).

Transcriptomic analysis further confirmed the upregulation of genes involved in ciliary assembly (*NEK2*), ciliary basal body (*CFAP36*), and motile cilia (*DNAH9* and *TTPP*) in WOI endometrial assembloids, with *NEK2* showing the most significant difference (Fig. 5B). Meanwhile, expressions of proteins related to ciliary assembly, ciliary basal body, and motile cilia were also increased in WOI assembloids, with the most notable differences observed in TBC1D1, IFT22, and IFT57 (ciliary assembly), and PJA2 (ciliary basal body) (Fig. 5C). Acetyl- α -tubulin (cilia marker (15)) were highly expressed in the WOI assembloids (Fig. 5D).

Single-cell transcriptome analysis revealed hormone treatment increased ciliated epithelium and decreased unciliated epithelium in SEC and WOI groups (Fig. 3E). Ciliated epithelium functioned in protein binding, cilium organization and assembly, while unciliated epithelium acted on actin cytoskeleton and translation (Fig. S2E~F). The WOI assembloids' ciliated epithelium regulated vasculature development and displayed higher transcriptional activity than the SEC group (Fig. 5E). Ciliated-unciliated epithelium transition occurred during the menstrual cycle, which was regulated by key genes, such as GAS5, JUN, RPL and RPS families (Fig. 5F, Fig.S2K~S2M). The implementation of aforementioned functions depended on its interaction with other cells. CellPhoneDB is a useful tool to investigate ligand-receptor interactions between the cells(43). Ciliated epithelium of WOI assembloids interacted with immune cells and secretory epithelium, showing enhanced invasion ability via CD74-COPA(44), and ROR2-WNT5A(45), which was validated by proximity ligation assay (PLA) (Fig. 5G~5H).

In summary, the WOI assembloids revealed accumulated ciliated epithelium's role in preparing the implantation window, with increased ciliary assembly and motility compared to SEC assembloids.

Receptive endometrial assembloids experienced epithelial-mesenchymal transition (EMT)

The WOI assembloids displayed upregulated cell differentiation not only at assembloid level (Fig. 3B) but also at cellular level, which is represented by EMT. EMT is a common and crucial biological event in the endometrium during the implantation window(46). During the EMT process, epithelial cells lose their epithelial characteristics while gaining migratory and invasive properties of fibroblasts. Synthetic analysis of the transcriptome and proteome revealed increased EMT in WOI assembloids (Fig. S3G).

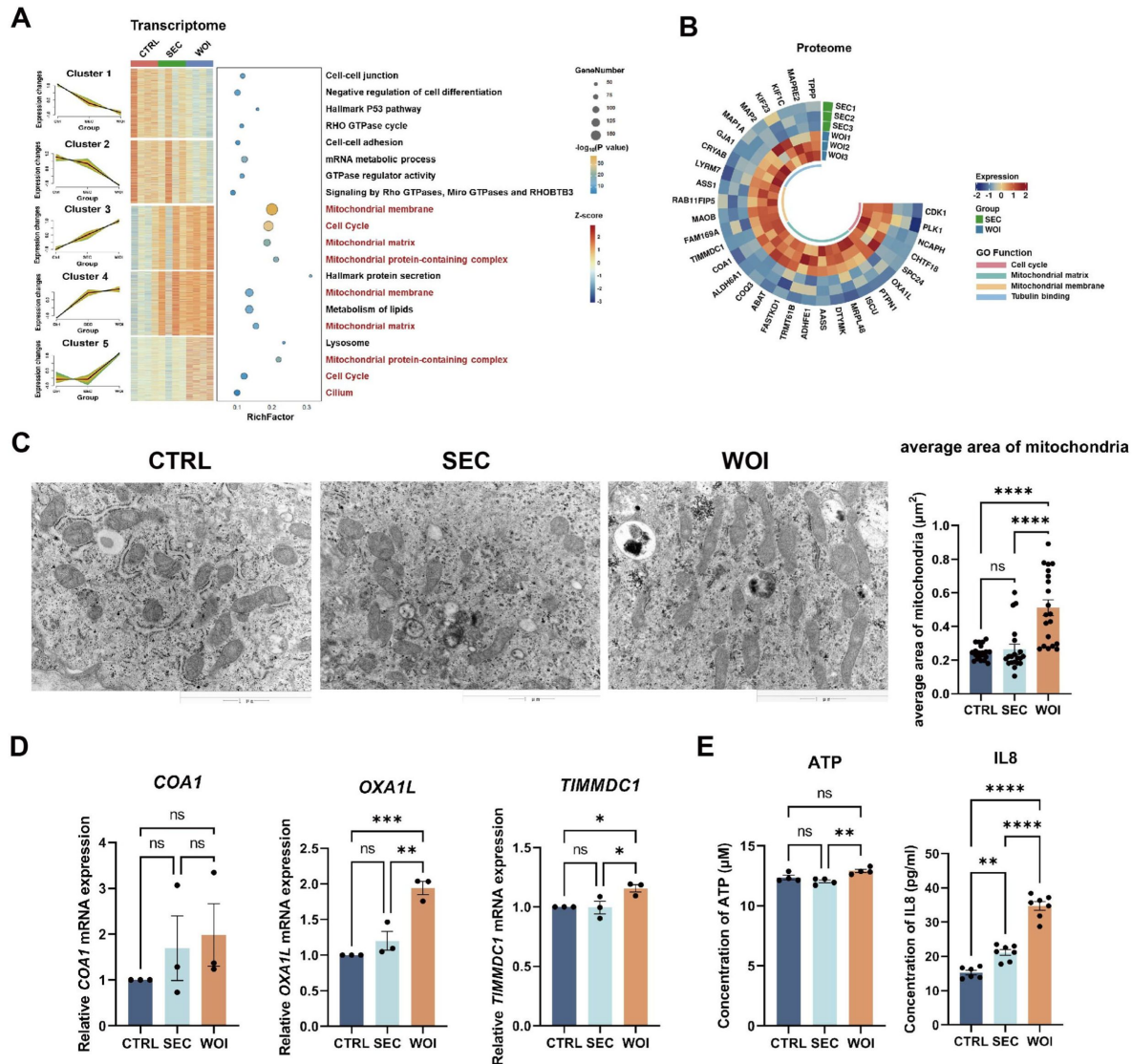


Fig 4

Receptive endometrial assembloids possess enhanced energy metabolism

(A) The Mfuzz trend analysis displayed the transcriptional variation trends of five clusters from CTRL, SEC, and WO1 groups (with a focus on the differences between SEC and WO1 assembloids). The heatmap illustrated corresponding gene expression profile (where color represents Z-score). The bubble plot showed the associated GO functions (with bubble size representing the number of genes and bubble color indicating the P value).

(B) The circular heatmap illustrated the functional differences of SEC and WO1 assembloids at the proteomic level. The color represents protein expression levels, and the innermost circle color represents GO functions.

(C) Transmission electron microscopy images displayed the mitochondrial morphology of CTRL, SEC, and WO1 assembloids, along with a quantitative comparison of mitochondrial area. Scale bar = 1 μm .

(D) RT-qPCR assessed the expression levels of mitochondrial function-related genes in the three assembloid groups.

(E) Quantitative comparison of the concentration of ATP (left) and IL8 (right) released by CTRL, SEC and WO1 assembloids.

* $P < 0.05$; ** $P < 0.005$; *** $P < 0.0005$; **** $P < 0.0001$.

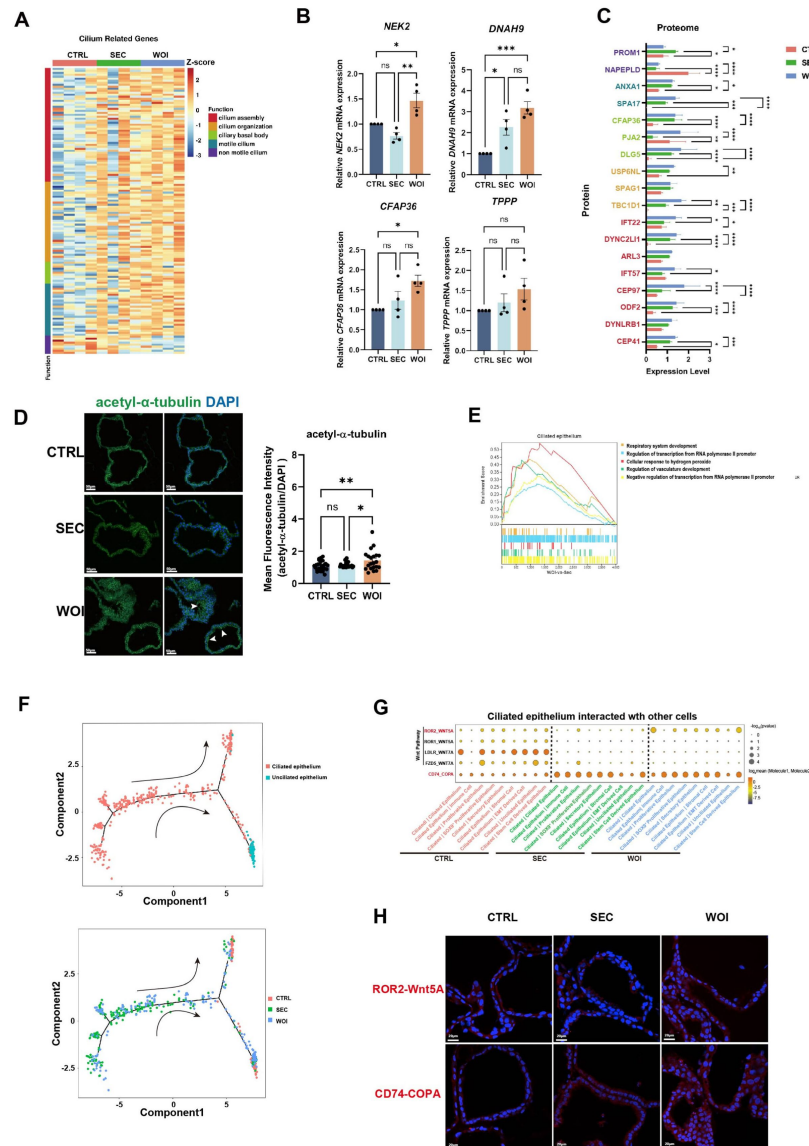


Fig 5

Receptive endometrial assembloids increased the ciliary assembly and motility

(A) The heatmap illustrated the expression of cilia-related genes across the CTRL, SEC, and WOI assembloids. The color represents Z-score, while the leftmost block indicates various characteristic functions related to cilia.

(B) RT-qPCR assessed the expression levels of cilia-related genes in the three assembloids.

(C) The histogram showed the expression of cilia-related proteins in the three groups of assembloids. The color of the longitudinal protein names corresponds to the color of cilia-related functional blocks in [Fig 5A](#).

(D) IF analysis of cilia assembly marked by acetyl-α-tubulin. Nuclei were counterstained with DAPI. The arrowhead indicates cilia. Scale bar = 50 μm.

(E) GSEA between the SEC and WOI groups for ciliated epithelium.

(F) Pseudotime trajectory showing the transformation between ciliated and unciliated epithelium in the CTRL, SEC and WOI groups. Arrows indicate the direction of the pseudotime trajectory.

(G) Dot plots demonstrating the Cellphone DB analysis of relevant receptors and ligands of ciliated epithelium with other cell types. The size of the dot represents the level of significance. The color of the dot indicates the mean of the average expression level of interacting molecule 1 in ciliated epithelium and molecule 2 in other cell types.

(H) Proximity ligation assay (PLA) validating the interactions of ROR2-Wnt5A and CD74-COPA in the CTRL, SEC and WOI assembloids. Red signals the interaction of two proteins. Nuclei were counterstained with DAPI. Scale bar = 20 μm.

* $P < 0.05$ ** $P < 0.005$ *** $P < 0.0005$ **** $P < 0.0001$.

EMT-derived cells, exhibiting gene expression patterns typical of epithelial and stromal cells, as well as EMT, are more abundant in the SEC and WOI groups, and act on protein binding, cell cycle, organelle organization, and reproduction (**Fig. S2E**). They performed enhanced lamellipodium-mediated cell migration, cell junction and cytoskeleton regulation in the WOI assembloids compared to the SEC assembloids (**Fig. S4A**). The WOI assembloids exhibited more thoroughly differentiation from proliferative epithelium to EMT-derived cells than the SEC assembloids, which was regulated by key genes such as *DOC2B*, *FXYD3*, and *LPCAT3* (**Fig. S4B** ~ **S4D**). EMT-derived cells and epithelium cooperated during the implantation window (**Fig. S4E**). We found that *NRP1* and *SLC7A1* were highly expressed by EMT-derived cells, and their receptors (*SEMA3A* and *CSF1*) were more upregulated in the epithelium of WOI group than SEC group (**Fig. S4E**, **S4G**). *NRP1*-*SEMA3A* has been reported to promote vascularization and responds to hypoxia(47). *SLC7A1*(48) and *CSF1*(49) both support receptivity establishment, embryo implantation and development. Compared with common stromal cells, EMT-derived cells communicate slightly differently with epithelial or immune cells. *CD44* and *CD46*, known for their roles in cell adhesion (51) and immunoregulation(52), are highly expressed in stromal cells and EMT-derived cells, respectively, and bind separately with *SPP1* and *JAG1* in epithelial and stromal cells (**Fig. S4E** ~ **S4G**).

In general, WOI assembloids demonstrated EMT's role in mediating endometrial transformation towards the implantation window.

The receptive endometrial assembloids possess the potential for embryo implantation

We further investigated the key biological function of the implantation window, specifically embryo implantation, using the assembloids. Given the rarity and ethical constraints associated with human embryos, we employed blastoids (corresponding to the human embryo at 6 days post-fertilization, referred to as “Day 6”) for implantation into the endometrial assembloids (**Fig.6A**). By Day 9, we observed that the blastoids could grow within the endometrial assembloids and interact with them (**Fig.6B**). The co-cultured blastoids displayed normal tri-lineage differentiation, characterized by markers for the epiblast (*OCT4*), hypoblast (*GATA6*), and trophoblast (*KRT18*) (**Fig.6C**). Additionally, we conducted a comparative analysis of the survival rates of blastoids and their interaction rates with endometrial assembloids across the CTRL, SEC and WOI assembloids. Remarkably, the survival and interaction rates of blastoids in the WOI endometrial assembloids were significantly higher compared to the CTRL and SEC groups, with survival rates of 66%, 19%, and 28%, and interaction rates of 90%, 47%, and 53% respectively, indicating the promising potential for embryo implantation (**Fig.6D~E**). This demonstrates that we have provided an innovative and supportive model suitable for embryo implantation.

In summary, WOI endometrial assembloids not only exhibited the typical structural and molecular features of the implantation window but also demonstrated significant potential for embryo implantation.

Discussion

In our study, we constructed the WOI endometrial assembloids, and observed the remarkable resemblance in structure and function to the *in vivo* endometrium. The assembloids consist of three primary types of endometrial cells, specifically epithelial, stromal, and immune cells, with epithelial cells assembling into glands and surrounded by immune cells and stromal cells. This similarity in cellular composition and tissue architecture to the endometrium lays the foundation for simulating receptive endometrium.

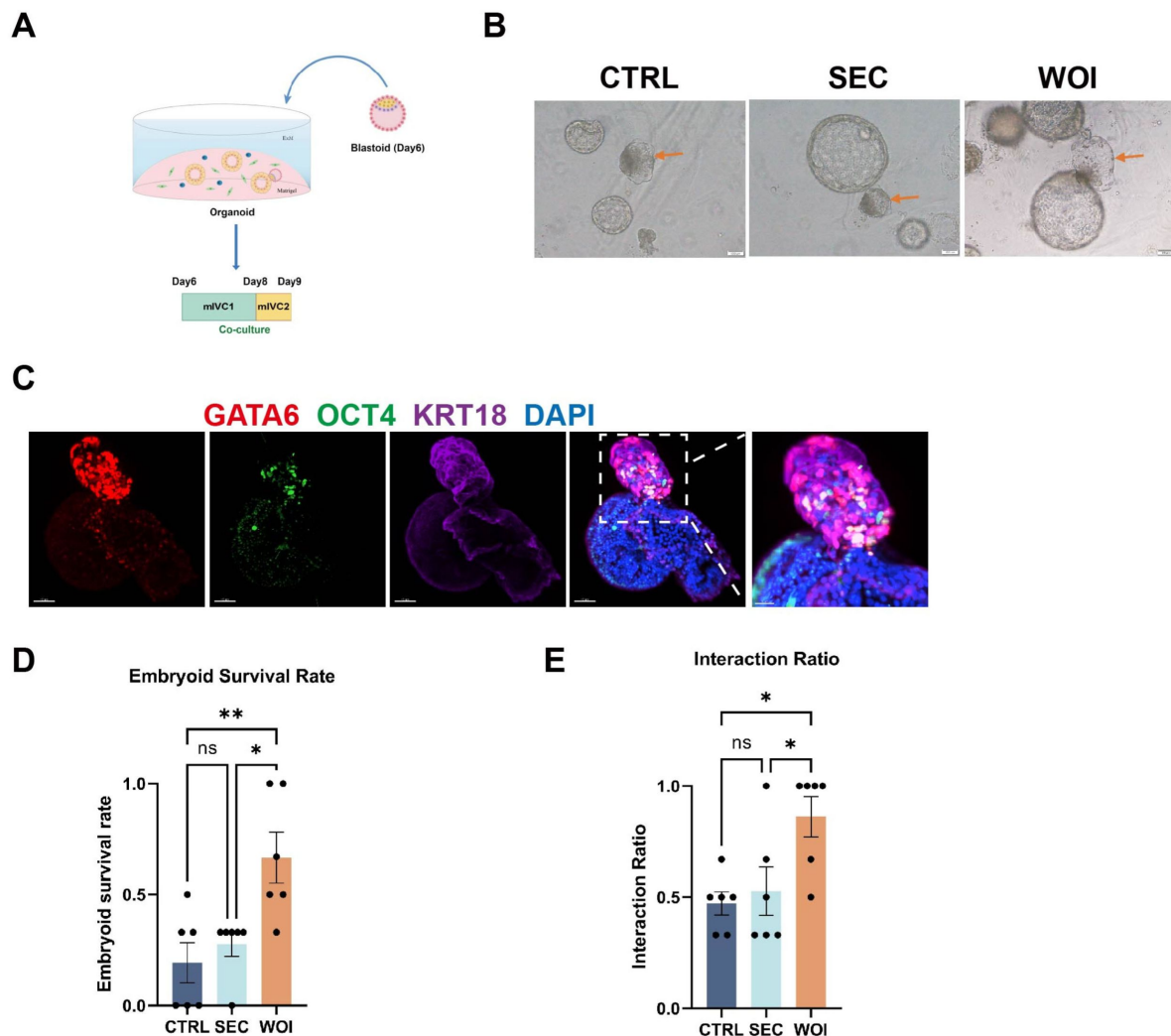


Fig 6

The receptive endometrial assembloids possess the potential for embryo implantation.

- (A) Diagram illustrated the co-culture model of endometrial assembloids with blastoids (the blastoid stage corresponds to a 6-day post-fertilization human embryo, referred to as Day 6 here).
- (B) Bright-field images of the co-culture of CTRL, SEC, and WOI assembloids with blastoids (Day 9) (yellow arrows indicate the blastoids). Scale bar = 100 μ m.
- (C) Whole-mount fluorescence staining of Day 9 co-cultured embryo and assembloid. OCT4 indicates the epiblast, GATA6 indicates the hypoblast, and KRT18 indicates the trophoblast. Scale bar = 40 μ m and 20 μ m (the rightmost image).
- (D) Comparison of the survival rates of Day 9 embryo in CTRL, SEC, and WOI assembloids.
- (E) Comparison of the interaction ratios between Day 9 embryo and endometrial assembloids in the CTRL, SEC, and WOI groups. * $P < 0.05$, ** $P < 0.005$.

Although previous studies treated endometrial assembloid with E2, P4, and cAMP to induce a transition to the secretory phase, the methodology to accurately simulate the implantation window or mid-secretory endometrium remains elusive and warrants further investigation. Previous research suggested that placental signals could further promote the differentiation of endometrial assembloids (2), and PRL, hCG and HPL have been implicated in processes such as decidualization, implantation, immunoregulation, and angiogenesis (21, 22, 24). Detailly, PRL, synthesized by the adenohypophysis, endometrium, and myometrium, plays a vital role in implantation, immunoregulation, and angiogenesis, with the secretory endometrium producing PRL in response to MPA and E2, leading to ciliated cell formation and stromal cell decidualization (2, 21). HCG, secreted by trophoblasts in early pregnancy, influences decidual cells (22) and improves endometrial thickness and receptivity(23). The introduction of hCG has been shown to elevate the expression of critical factors associated with endometrial receptivity, such as endocytosis proteins, hypoxia-inducible factor 1 (HIF1), chemokines, and glycodelin (53). HPL contributes to the development and function of uterine glands(24). Consequently, the additional supplementation of PRL, hCG and HPL in our system augmented hormone responsiveness and receptivity, while promoting cell differentiation, ultimately yielding a model more representative of the receptive endometrium.

Here, the WOI assembloid exhibited the characteristic ultra-structures, such as cilia. During the human menstrual cycle, motile cilia are present in the epithelium of the endometrium(1). These hair-like organelles extend from the cell surface and beat rhythmically, facilitating cell and tissue movement while driving fluid transport across the epithelium (54). During the decidualization of the endometrium, the number and length of cilia increase, a process driven by the two primary regulatory factors for embryo implantation: estrogen and progesterone (15, 55, 56). However, with aging, the expression of cilia-related genes in the endometrium is downregulated(57). In patients with recurrent implantation failure (RIF), ciliary defects in the endometrium are observed, leading to abnormal decidualization and repeated implantation failure(56). Thus, cilia play a crucial role in endometrial decidualization and embryo implantation. Consistently, the WOI assembloid exhibited increased ciliated epithelium, along with enhanced assembly and motility of cilia.

Furthermore, the WOI assembloid demonstrated energy and lipid metabolism patterns resembling those of the in vivo receptive endometrium. Enhanced energy metabolism involves monocarboxylic acid metabolism and mitochondrial oxidative phosphorylation. Monocarboxylic acids, exemplified by lactate, pyruvate, and ketone bodies, are vital metabolites in most mammalian cells. During the implantation window, elevated lactate levels mobilize endometrial monocarboxylic acid metabolism and function as a pregnancy-related signal, stimulating secretion in the epithelium. Subsequently, ATP, the primary product of energy metabolism, induces neighboring epithelial cells to release IL8, promoting decidualization of stromal cells (42). It was also observed that WOI assembloids indeed possessed larger and functionally active mitochondria, producing much more ATP and IL8 than CTRL and SEC assembloids. Lipid metabolism, responsible for energy storage, signal transduction, cell proliferation, apoptosis, and membrane trafficking, plays a crucial role in endometrial receptivity and implantation, although the precise mechanisms are not fully understood (58–60). Thus, the WOI assembloids possessed metabolic characteristics of in vivo implantation window.

However, our WOI endometrial assembloids also exhibit some limitations. It is undeniable that the assembloids cannot perfectly replicate the in vivo endometrium, which comprises functional and basal layers with a greater abundance of cell subtypes, under superior regulation by systemic hormones and nutrients. Stromal cells and immune cells are difficult to pass down stably and their proportion is lower than that in the in vivo endometrium. We are looking forward to combining stem cell induction, 3D printing technology, and microfluidic systems to alter the culture environment in order to optimize the existing model.

In summary, we developed a receptive endometrial assembloid that mimics the features of in vivo implantation-window endometrium (**Fig. 7**). With typical ultra-structures of the implantation window, including pinopodes and cilia, it not only demonstrated hormonal responsiveness and characteristic glycogen secretion functions, but also recapitulated the processes such as decidualization, metabolic changes and EMT, while retaining the potential for embryo implantation. This receptive endometrial assembloid serves as a platform to investigate peri-implantation endometrial physiology and pathology, maternal-fetal interactions, with potential practical applications and clinical translation.

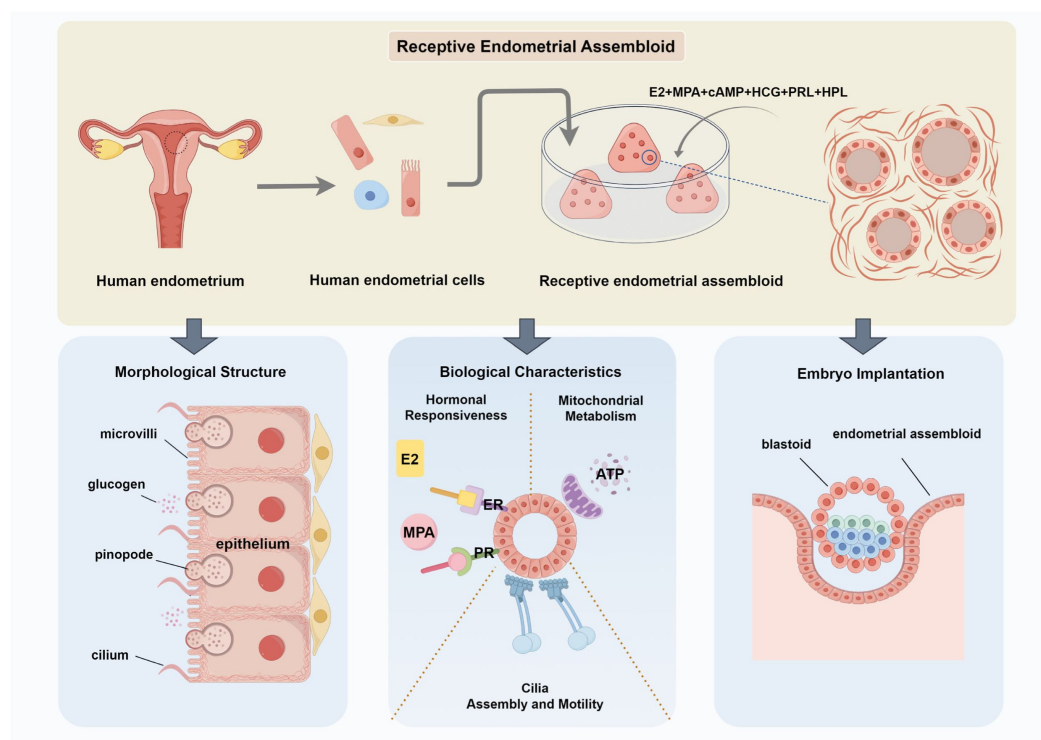


Fig.7

Schematic diagram displaying the establishment and validation of receptive endometrial assembloids, and summarizing the characteristic biological events of implantation-window endometrium.

Materials and Methods

Establishment of endometrial assembloids

All experiments involving human subjects followed medical ethical principles and the Declaration of Helsinki, and were approved by the Ethics Committee of Qilu Hospital of Shandong University. Informed consents were obtained from patients. Females of reproductive age who received hysterectomy for benign diseases were selected for this study. Clinical information of patients providing endometrial tissue was listed in Dataset S1. Experimental procedures are presented in detail in the Supporting Information.

Hormone treatment of endometrial assembloids

The hormone regimen for inducing endometrial secretory phase, as described by Thomas E. Spencer et al (9 [↗](#)), consisted of estradiol for 2 days followed by a combination of estradiol (Sigma E2758), medroxyprogesterone acetate (Selleck S2567) and N6,2'-O-dibutyryladenosine 3',5'-cyclic monophosphate sodium salt (cAMP) (Sigma D0627) for 6 days.

To simulate the endometrium of implantation window, we developed a model by incorporating various pregnancy-related hormones. The hormone regimen included estradiol for the first two days, followed by a combination of estradiol, medroxyprogesterone acetate, cAMP, Human Chorionic Gonadotropin (HCG) (Livzon Pharmaceutical Group Inc), Human Placental Lactogen (HPL) (R&D Systems 5757-PL), and prolactin (Peprotech 100-07) for 6 days. The CTRL group was cultured in ExM without the addition of hormones. (**Fig. 1A** [↗](#)) (Table S2)

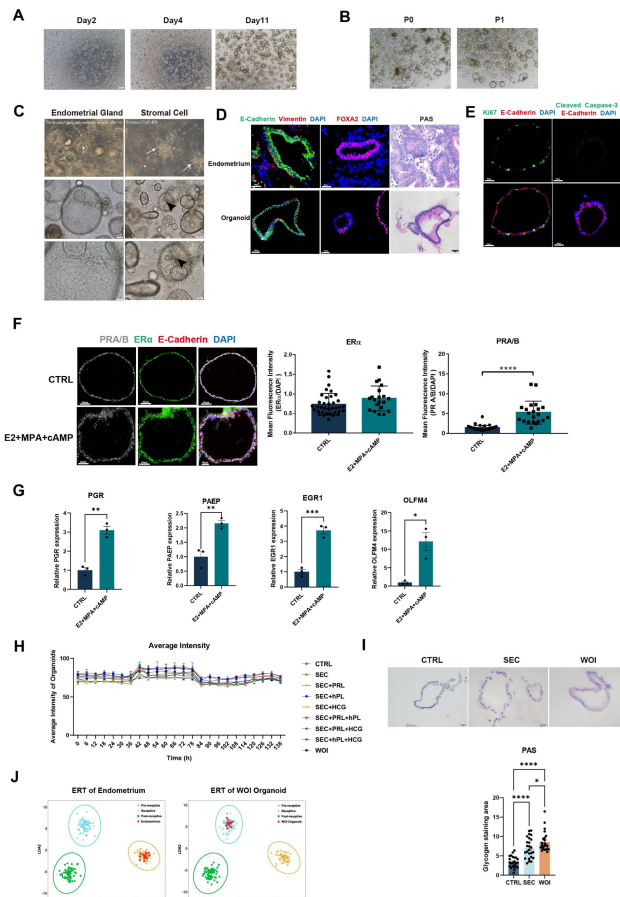
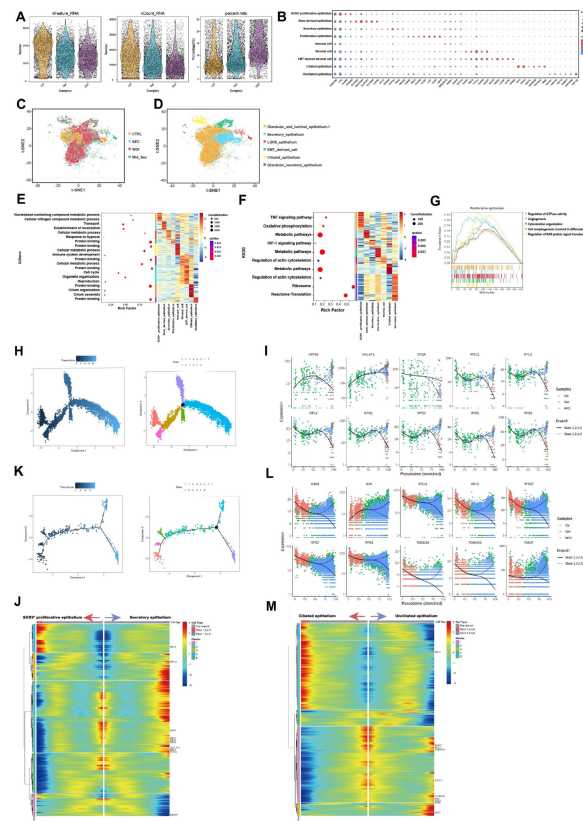


Fig. S1

Developing receptive endometrial assembloids in vitro

- (A) Brightfield of endometrial assembloids on day2, day4 and day11. Scale bar = 200 μ m.
- (B) Brightfield of endometrial assembloids in the primary generation (P0) and first generation (P1). Scale bar = 200 μ m.
- (C) Screenshot of video S1 showing endometrial glands gradually developing into a vesicular shape, and the surrounding stromal cells arranging in a fibrous pattern in the CTRL endometrial assembloids (100x, up-left) (The yellow arrows indicate stromal cells and the white arrows indicate endometrial glands). Screenshot of video S2 displaying the stromal cells growing in fibrous pattern and forming an extensive network in the CTRL endometrial assembloids (200x, up-right) (The white arrows indicate stromal cells). The epithelial cells arrange like paving stones (middle and down, left). Stromal cells formed an extensive network (middle and down, right) (The arrowhead indicates stromal cells). Scale bar = 100 μ m (middle), Scale bar = 50 μ m (down).
- (D) Validation of epithelial, stromal cell and endometrial gland markers (E-cadherin, vimentin and FOXA2, respectively) with immunofluorescence (IF) in the endometrium in vivo and CTRL endometrial assembloids in vitro. Nuclei were counterstained with DAPI. Scale bar = 40 μ m. Periodic acid-schiff staining (PAS) of endometrium in vivo and endometrial assembloid in vitro. Scale bar = 20 μ m.
- (E) IF analysis of proliferation and apoptosis indicated by Ki67 and cleaved caspase-3 in the CTRL assembloids, respectively. Nuclei were counterstained with DAPI. Scale bar = 40 μ m.
- (F) Verification of hormone responsiveness by the expression levels of ER α and PRA/B (E-cadherin indicated the marker of epithelial cell). Scale bar = 40 μ m. *** $P \leq 0.0005$.
- (G) Relative expression of PGR, PAEP, EGR1, and OLFM4 in the CTRL and hormone-treated assembloids by RT-qPCR. * $P \leq 0.05$, ** $P \leq 0.005$, *** $P \leq 0.0005$.
- (H) The dynamic changes of the average intensity of assembloids over time in each hormone regimen.
- (I) PAS of CTRL, SEC and WOI assembloids. Scale bar = 20 μ m. Quantitative comparison of glycogen staining area in the CTRL, SEC and WOI assembloids. * $P \leq 0.05$, *** $P \leq 0.0001$.
- (J) Endometrial receptivity evaluation of endometrium and their derived WOI assembloids through ERT. Asterisks indicate individual samples.



FigS2

Various functions performed by all kinds of cells identified with scRNA-seq.

(A) Box plot of the gene numbers detected, UMI numbers, and ratios of mitochondrial gene expression in single cells of the CTRL, SEC and WOI groups.

(B) Bubble diagram showing the distribution of marker gene expression in each cluster.

(C~D) Comparison of cell composition between CTRL, SEC, WOI endometrial assembloids and mid-secretory endometrium demonstrating samples (C) and cell types (D).

(E) Bubble diagram and heatmap showing the corresponding upregulated genes and GO function of each cluster of endometrial cells.

(F) Bubble diagram and heatmap showing corresponding upregulated genes and KEGG functions of SOX9⁺ proliferative epithelium, stem-derived epithelium, secretory epithelium, proliferative epithelium, stromal cells, ciliated epithelium and unciliated epithelium. Color is proportional to log-transformed fold change of gene expression.

(G) GSEA between the SEC and WOI groups for proliferative epithelium.

(H) The single-cell pseudotime trajectory of SOX9⁺ proliferative epithelium, secretory epithelium and proliferative epithelium. Cells start at proliferative epithelium and progress to SOX9⁺ proliferative epithelium and secretory epithelium (left). There are seven major states over pseudotime (right). The black spot indicates the differentiation node between state 5 and state 6, indicating the direction from proliferative epithelium to SOX9⁺ proliferative epithelium and secretory epithelium, respectively.

(I)(L) The horizontal axis is the pseudotime point, and the vertical axis is the gene expression level. The solid line represents states 1, 2, 4, and 5 corresponding to Fig. S2H (I) or Fig. S2K (L). Different colors represent samples in the CTRL, SEC and WOI groups.

(J)(M) Heatmap of genes at the branch node regulating differentiation into SOX9⁺ proliferative epithelium or secretory epithelium (J), and ciliated epithelium or unciliated epithelium (M). The horizontal axis is the pseudo-time point (the pseudo-time point gradually increases from the middle to both sides). The vertical axis is the gene expression level, representing two differential directions on the left and right sides. Clusters represent the gene sets with a similar branch gene expression trend. Different colors represent the level of gene expression.

(K) Pseudotime trajectory of ciliated and unciliated epithelium. Cells start at ciliated epithelium and progress to unciliated epithelium (left). There are seven major states over pseudotime (right). The black spot indicates the differentiation node between state 5 and state 6, indicating the direction of ciliated and unciliated epithelium, respectively. Arrows indicate the direction of the pseudotime trajectory.

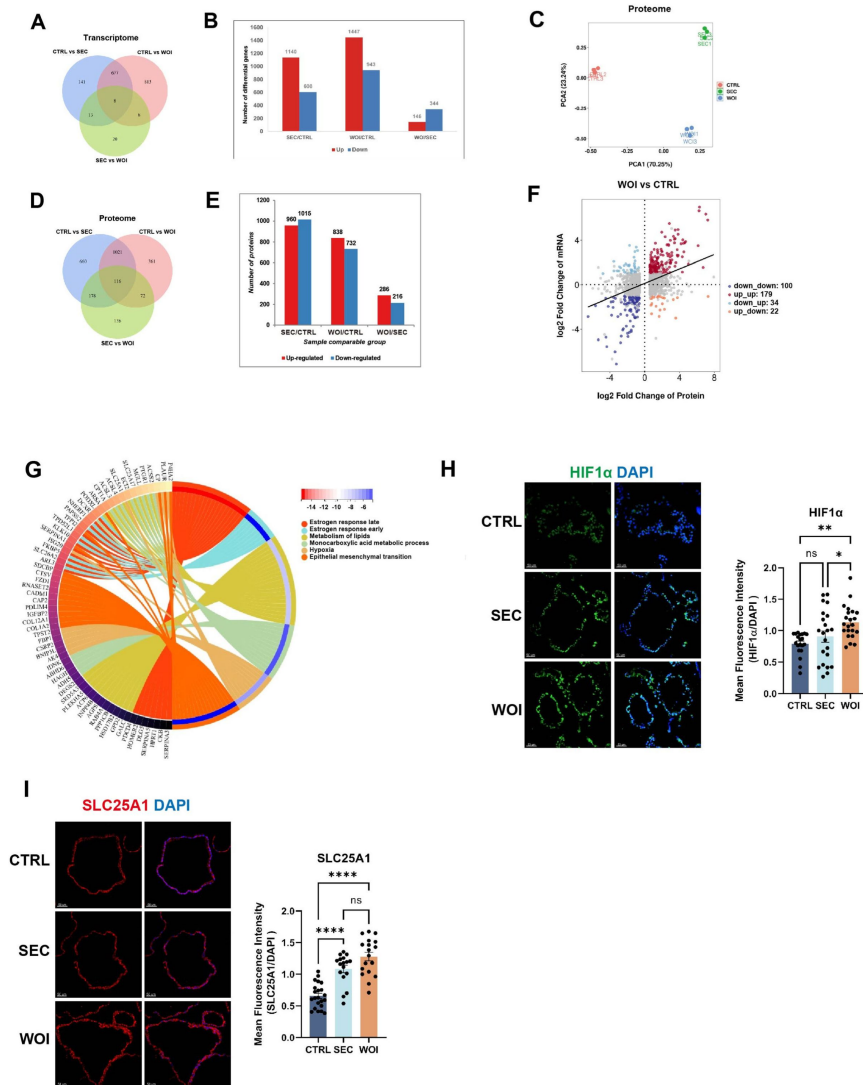


Fig S3

Comparisons between CTRL, SEC and WOI assembloids at the level of transcriptome and proteome.

- (A) Venn diagram showing differential genes in pairs among the three groups. Log₂ FC (Fold Change) > 1.2 or < -1.2, q value < 0.05.
- (B) Histogram showing pairwise comparison of differential gene expression (DEG) from bulk RNA-seq among the CTRL, SEC and WOI groups. qvalue < 0.05, |logFC| > 1.
- (C) PCA plot computed with differentially expressed proteins in the micro proteomics of endometrial assembloids belonging to the CTRL, SEC and WOI groups.
- (D) Venn diagram showing differential proteins in pairs among the three groups. Log₂ FC > 1.2 or < -1.2, q value < 0.05.
- (E) Histogram showing pairwise comparison of differential proteins in micro proteomics among the CTRL, SEC and WOI groups.
- (F) Scatterplot depicting the correlation between the transcriptome and proteome as for the WOI group and CTRL group. For transcriptome, FC > 2 and p value < 0.05 were defined as significantly upregulated. For proteome, FC > 1.2 and p value < 0.05 were defined as significantly upregulated.
- (G) Circle diagram showing the functions of genes upregulated in both the transcriptome and proteome as for the WOI group compared to CTRL group.
- (H) Verification of hypoxia response by the expression levels of HIF1α. Scale bar = 50 μm. *P ≤ 0.05, **P ≤ 0.005.
- (I) Verification of lipid metabolism by the expression levels of SLC25A1. Scale bar = 50 μm. ****P ≤ 0.0001.

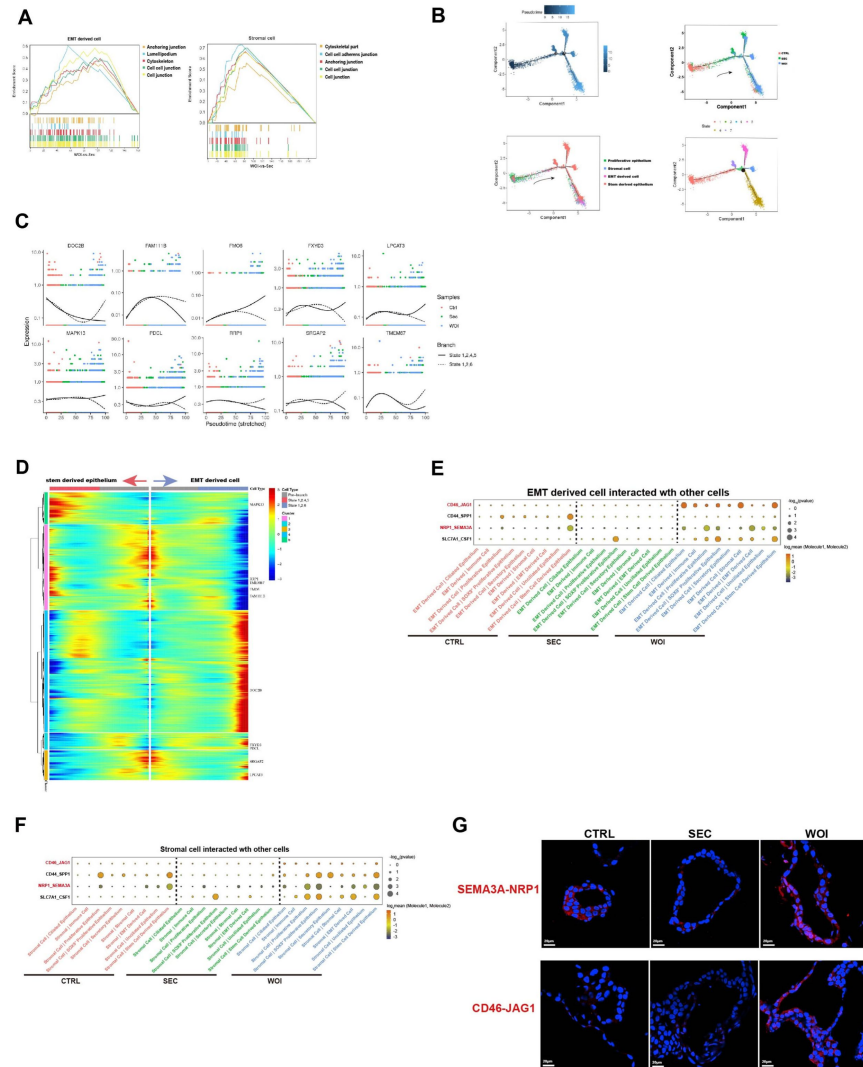


Fig S4

Receptive endometrial assembloids experienced epithelial-mesenchymal transition (EMT)

(A) GSEA between the SEC and WOI groups for proliferative epithelium.

(B) The single-cell pseudotime trajectory of proliferative epithelium, stromal cell, EMT derived cell and stem derived epithelium. Cells start at proliferative epithelium and progress to EMT derived cell. There are seven major states over pseudotime. The black spot indicates the differentiation node between state 4, 5 and state 6, indicating the direction from proliferative epithelium to EMT derived cell. Arrows indicate the direction of the pseudotime trajectory.

(C) The horizontal axis is the pseudotime point, and the vertical axis is the gene expression level. The solid line represents states 1, 2, 4, and 5 corresponding to Fig. S4B. Different colors represent samples in the CTRL, SEC and WOI groups.

(D) Heatmap of genes at the branch node regulating differentiation into EMT derived cell and stem-derived epithelium. The horizontal axis is the pseudo-time point (the pseudo-time point gradually increases from the middle to both sides). The vertical axis is the gene expression level, representing two differential directions on the left and right sides. Clusters represent the gene sets with a similar branch gene expression trend. Different colors represent the level of gene expression.

(E-F) Dot plots demonstrating the Cellphone DB analysis of relevant receptors and ligands of EMT derived cell (E) or stromal cell (F) with other cell types. The size of the dot represents the level of significance. The color of the dot indicates the mean of the average expression level of interacting molecule 1 in EMT derived cells (E) or stromal cells (F) and molecule 2 in other cell types.

(H) Proximity ligation assay (PLA) validating the interactions of SEMA3A-NRP1 and CD46-JAG1 in the CTRL, SEC and WOI assembloids. Red signals the interaction of two proteins. Nuclei were counterstained with DAPI. Scale bar = 20 μ m.

Acknowledgements

This work was supported by National Natural Science Foundation of China (82192874 to H.Z), the National Key Research and Development Program of China (2021YFC2700301 to K.W), the Basic Science Center Program (31988101 to Z.-J.C.), Fundamental Research Funds for the Central Universities(2022JC006 to K.W), the Fundamental Research Funds of Shandong University Taishan Scholars Program of Shandong Province (ts20190988 to H.Z.; tsqn201909194 to K.W), Innovative research team of high-level local universities in Shanghai (SHSMU-ZLCX20210201 to K.W).

We sincerely appreciate Prof. Tianqing Li (Kunming University of Science and Technology) and Prof. Shaorong Gao (Tongji University) for insightful comments and discussions.

We are grateful to Guangzhou Genedenovo Biotechnology Co., Ltd for assisting in sequencing and bioinformatics analysis. We also thank to YiKon Medical from China for technological assistance of ERT, Jingjie PTM Biolab (HangZhou) Co., Inc. for proteomics.

Additional information

Author contributions

H. Zhao, K. Wu and Z.-J.C conceived the project and supervised research. Y. Zhang, R. Zhao and H. Zhao designed the experiments. Y. Zhang and R. Zhao performed the experiments, analyzed and interpreted data. Y. Zhang wrote the manuscript. R. Zhao, Y. Li, B. Liu, and H. Zhao revised the manuscript. P. Liu provided human endometrium samples. C.Yang, J. Song, T. Li, C. Yin, M. Lu, Z. Hou and C. Zhang provided technical support. All authors revised and approved the manuscript.

Additional files

supporting information [↗](#)

Video_Table_Dateset [↗](#)

VideoS1 [↗](#)

VideoS2 [↗](#)

VideoS3 [↗](#)

VideoS4 [↗](#)

VideoS5 [↗](#)

dataset [↗](#)

References

1. Wang W, et al. (2020) **Single-cell transcriptomic atlas of the human endometrium during the menstrual cycle** *Nat Med* **26**:1644–1653 [Google Scholar](#)
2. Turco MY, et al. (2017) **Long-term, hormone-responsive organoid cultures of human endometrium in a chemically defined medium** *Nat Cell Biol* **19**:568–577 [Google Scholar](#)
3. Boretto M, et al. (2017) **Development of organoids from mouse and human endometrium showing endometrial epithelium physiology and long-term expandability** *Development* **144**:1775–1786 [Google Scholar](#)
4. Cindrova-Davies T, et al. (2021) **Menstrual flow as a non-invasive source of endometrial organoids** *Commun Biol* **4**:651 [Google Scholar](#)
5. Shun Shibata SE, Nagai Luis A. E., Kobayashi Eri H., Oike Akira, Kobayashi Norio, Kitamura Akane, Hori Takeshi, Nashimoto Yuji, Nakato Ryuichiro, Hamada Hirotake, Kaji Hirokazu, Kikutake Chie, Suyama Mikita, Saito Masatoshi, Yaegashi HO Nobuo, Arima Takahiro (2024) **Modeling embryo-endometrial interface recapitulating human embryo implantation** *Sci Adv* **10**:ead4819 [Google Scholar](#)
6. Cheung VC, et al. (2021) **Pluripotent stem cell-derived endometrial stromal fibroblasts in a cyclic, hormone-responsive, coculture model of human decidua** *Cell Rep* **35**:109138 [Google Scholar](#)
7. Gong L, et al. (2022) **Bi-potential hPSC-derived Mullerian duct-like cells for full-thickness and functional endometrium regeneration** *NPJ Regen Med* **7**:68 [Google Scholar](#)
8. Dolat L, Valdivia RH (2021) **An endometrial organoid model of interactions between Chlamydia and epithelial and immune cells** *J Cell Sci* **134** [Google Scholar](#)
9. Harriet C, Fitzgerald PD, Behura Susanta K., Schust Danny J., Spencer Thomas E. (2019) **Self-renewing endometrial epithelial organoids of the human uterus** *Proc Natl Acad Sci U S A* **116**:23132–23142 [Google Scholar](#)
10. Wiwatpanit T, et al. (2020) **Scaffold-Free Endometrial Organoids Respond to Excess Androgens Associated With Polycystic Ovarian Syndrome** *J Clin Endocrinol Metab* **105**:769–780 [Google Scholar](#)
11. Thomas M, Rawlings KM, Taylor Deborah M, Molè Matteo A, Fishwick Katherine J, Tryfonos Maria, Odendaal Joshua, Hawkes Amelia, Zernicka-Goetz Magdalena, Hartshorne Geraldine M, Brosens Jan J, Lucas Emma S (2021) **Modelling the impact of decidual senescence on embryo implantation in human endometrial assembloids** *eLife* **10**:e69603 <https://doi.org/10.7554/eLife.69603> | [Google Scholar](#)
12. Park SR, et al. (2021) **3D stem cell-laden artificial endometrium: successful endometrial regeneration and pregnancy** *Biofabrication* **13** [Google Scholar](#)
13. Park SR, et al. (2023) **Development of cell-laden multimodular Lego-like customizable endometrial tissue assembly for successful tissue regeneration** *Biomater Res* **27**:33 [Google](#)

Scholar

14. Boretto M, et al. (2019) **Patient-derived organoids from endometrial disease capture clinical heterogeneity and are amenable to drug screening** *Nat Cell Biol* **21**:1041–1051 [Google Scholar](#)
15. Haider S, et al. (2019) **Estrogen Signaling Drives Ciliogenesis in Human Endometrial Organoids** *Endocrinology* **160**:2282–2297 [Google Scholar](#)
16. Cochrane DR, et al. (2020) **Single cell transcriptomes of normal endometrial derived organoids uncover novel cell type markers and cryptic differentiation of primary tumours** *J Pathol* **252**:201–214 [Google Scholar](#)
17. Bui BN, et al. (2020) **Organoids can be established reliably from cryopreserved biopsy catheter-derived endometrial tissue of infertile women** *Reprod Biomed Online* **41**:465–473 [Google Scholar](#)
18. Fitzgerald HC, Kelleher AM, Ranjit C, Schust DJ, Spencer TE (2023) **Basolateral secretions of human endometrial epithelial organoids impact stromal cell decidualization** *Mol Hum Reprod* **29** [Google Scholar](#)
19. Kirkwood PM, et al. (2022) **Single-cell RNA sequencing and lineage tracing confirm mesenchyme to epithelial transformation (MET) contributes to repair of the endometrium at menstruation** *eLife* **11** [Google Scholar](#)
20. Paule SG, et al. (2021) **Podocalyxin is a key negative regulator of human endometrial epithelial receptivity for embryo implantation** *Hum Reprod* **36**:1353–1366 [Google Scholar](#)
21. Auriemma RS, et al. (2020) **The Interplay Between Prolactin and Reproductive System: Focus on Uterine Pathophysiology** *Front Endocrinol* **11**:594370 [Google Scholar](#)
22. d'Hauterive SP, et al. (2022) **Human Chorionic Gonadotropin and Early Embryogenesis: Review** *Int J Mol Sci* **23** [Google Scholar](#)
23. Nwabuobi C, et al. (2017) **hCG: Biological Functions and Clinical Applications** *Int J Mol Sci* **18** [Google Scholar](#)
24. Bazer FW, Burghardt RC, Johnson GA, Spencer TE, Wu G (2018) **Mechanisms for the establishment and maintenance of pregnancy: synergies from scientific collaborations** *Biol Reprod* **99**:225–241 [Google Scholar](#)
25. Garcia-Alonso L, et al. (2021) **Mapping the temporal and spatial dynamics of the human endometrium in vivo and in vitro** *Nat Genet* **53**:1698–1711 [Google Scholar](#)
26. Rocío Martínez-Aguilar LEK, Reavey Jane J, Hilary O, Critchley D, Jacqueline A Maybin (2021) **HYPOXIA AND REPRODUCTIVE HEALTH: The presence and role of hypoxia in the endometrium** *Reproduction* **161**:F1–F17 [Google Scholar](#)
27. Fatmous M, Rai A, Poh QH, Salamonsen LA, Greening DW (2022) **Endometrial small extracellular vesicles regulate human trophoblastic cell invasion by reprogramming the phosphoproteome landscape** *Front Cell Dev Biol* **10**:1078096 [Google Scholar](#)
28. Jiang X, et al. (2022) **Lack of VMP1 impairs hepatic lipoprotein secretion and promotes non-alcoholic steatohepatitis** *J Hepatol* **77**:619–631 [Google Scholar](#)

29. Kuo WT, et al. (2021) **The Tight Junction Protein ZO-1 Is Dispensable for Barrier Function but Critical for Effective Mucosal Repair** *Gastroenterology* **161**:1924–1939 [Google Scholar](#)
30. Bissonnette L, et al. (2016) **Human S100A10 plays a crucial role in the acquisition of the endometrial receptivity phenotype** *Cell Adh Migr* **10**:282–298 [Google Scholar](#)
31. Srivastava C, et al. (2018) **FAT1 modulates EMT and stemness genes expression in hypoxic glioblastoma** *Int J Cancer* **142**:805–812 [Google Scholar](#)
32. Liu L, et al. (2020) **si-SNHG5-FOXF2 inhibits TGF-beta1-induced fibrosis in human primary endometrial stromal cells by the Wnt/beta-catenin signalling pathway** *Stem Cell Res Ther* **11**:479 [Google Scholar](#)
33. Geng J, Cui C, Yin Y, Zhao Y, Zhang C (2022) **LncRNA NEAT1 affects endometrial receptivity by regulating HOXA10 promoter activity** *Cell Cycle* **21**:1932–1944 [Google Scholar](#)
34. Vargas MF, et al. (2012) **Effect of single post-ovulatory administration of levonorgestrel on gene expression profile during the receptive period of the human endometrium** *J Mol Endocrinol* **48**:25–36 [Google Scholar](#)
35. Kinnear S, Salamonsen LA, Francois M, Harley V, Evans J (2019) **Uterine SOX17: a key player in human endometrial receptivity and embryo implantation** *Sci Rep* **9**:15495 [Google Scholar](#)
36. Huang P, et al. (2021) **SOX4 facilitates PGR protein stability and FOXO1 expression conducive for human endometrial decidualization** *eLife* **11**:e72073 <https://doi.org/10.7554/eLife.72073> | [Google Scholar](#)
37. He A, et al. (2021) **The role of transcriptomic biomarkers of endometrial receptivity in personalized embryo transfer for patients with repeated implantation failure** *J Transl Med* **19**:176 [Google Scholar](#)
38. Yang Y, et al. (2021) **SLC25A1 promotes tumor growth and survival by reprogramming energy metabolism in colorectal cancer** *Cell Death Dis* **12**:1108 [Google Scholar](#)
39. Wang C, et al. (2019) **MITRAC15/COA1 promotes mitochondrial translation in a ND2 ribosome–nascent chain complex** *EMBO reports* **21** [Google Scholar](#)
40. Itoh Y, et al. (2021) **Mechanism of membrane-tethered mitochondrial protein synthesis** *Science* **371**:846–849 [Google Scholar](#)
41. Wang T, et al. (2021) **C9orf72 regulates energy homeostasis by stabilizing mitochondrial complex I assembly** *Cell Metabolism* **33**:531–546 [Google Scholar](#)
42. Gu XW, et al. (2020) **ATP mediates the interaction between human blastocyst and endometrium** *Cell Prolif* **53**:e12737 [Google Scholar](#)
43. Efremova M, Vento-Tormo M, Teichmann SA, Vento-Tormo R (2020) **CellPhoneDB: inferring cell-cell communication from combined expression of multi-subunit ligand-receptor complexes** *Nat Protoc* **15**:1484–1506 [Google Scholar](#)
44. Li M, et al. (2022) **BUB1 Is Identified as a Potential Therapeutic Target for Pancreatic Cancer Treatment** *Front Public Health* **10**:900853 [Google Scholar](#)

45. Li T, et al. (2022) **WNT5A Interacts With FZD5 and LRP5 to Regulate Proliferation and Self-Renewal of Endometrial Mesenchymal Stem-Like Cells** *Front Cell Dev Biol* **10**:837827 [Google Scholar](#)
46. Owusu-Akyaw A, Krishnamoorthy K, Goldsmith LT, Morelli SS (2019) **The role of mesenchymal-epithelial transition in endometrial function** *Hum Reprod Update* **25**:114–133 [Google Scholar](#)
47. Casazza A, et al. (2013) **Impeding macrophage entry into hypoxic tumor areas by Sema3A/Nrp1 signaling blockade inhibits angiogenesis and restores antitumor immunity** *Cancer Cell* **24**:695–709 [Google Scholar](#)
48. Wang X, et al. (2014) **Functional role of arginine during the peri-implantation period of pregnancy. I. Consequences of loss of function of arginine transporter SLC7A1 mRNA in ovine conceptus trophoctoderm** *FASEB J* **28**:2852–2863 [Google Scholar](#)
49. Robertson SA, Chin PY, Femia JG, Brown HM (2018) **Embryotoxic cytokines-Potential roles in embryo loss and fetal programming** *J Reprod Immunol* **125**:80–88 [Google Scholar](#)
50. Du L, et al. (2021) **Single-cell transcriptome analysis reveals defective decidua stromal niche attributes to recurrent spontaneous abortion** *Cell Prolif* **54**:e13125 [Google Scholar](#)
51. Sancakli Usta C, Turan G, Bulbul CB, Usta A, Adali E (2020) **Differential expression of Oct-4, CD44, and E-cadherin in eutopic and ectopic endometrium in ovarian endometriomas and their correlations with clinicopathological variables** *Reprod Biol Endocrinol* **18**:116 [Google Scholar](#)
52. Le Friec G, et al. (2012) **The CD46-Jagged1 interaction is critical for human TH1 immunity** *Nat Immunol* **13**:1213–1221 [Google Scholar](#)
53. Bielfeld AP, et al. (2019) **A Proteome Approach Reveals Differences between Fertile Women and Patients with Repeated Implantation Failure on Endometrial Level(-)Does hCG Render the Endometrium of RIF Patients?** *Int J Mol Sci* **20** [Google Scholar](#)
54. Zhou Feng, Roy S (2015) **SnapShot: Motile Cilia** *Cell* **162**:224–224 [Google Scholar](#)
55. Li Y-PY Bo, He Yu-Ying, Liang Chen, Li Meng-Yuan, Wang Ying, Yang Zeng-Ming (2023) **IHH, SHH, and primary cilia mediate epithelial-stromal cross-talk during decidualization in mice** *Sci Signal* **16**:eadd0645 [Google Scholar](#)
56. Li B, et al. (2022) **Primary Cilia Restrain PI3K-AKT Signaling to Orchestrate Human Decidualization** *International Journal of Molecular Sciences* **23**:15573 [Google Scholar](#)
57. Devesa-Peiro A, Sebastian-Leon P, Parraga-Leon A, Pellicer A, Diaz-Gimeno P (2022) **Breaking the ageing paradigm in endometrium: endometrial gene expression related to cilia and ageing hallmarks in women over 35 years** *Hum Reprod* **37**:762–776 [Google Scholar](#)
58. King K, et al. (2021) **Dynamics of lipid droplets in the endometrium and fatty acids and oxylipins in the uterine lumen, blood, and milk of lactating cows during diestrus** *J Dairy Sci* **104**:3676–3692 [Google Scholar](#)

59. Vilella F, Ramirez LB, Simon C (2013) **Lipidomics as an emerging tool to predict endometrial receptivity** *Fertil Steril* **99**:1100–1106 [Google Scholar](#)
60. Braga D, et al. (2019) **Lipidomic profile as a noninvasive tool to predict endometrial receptivity** *Mol Reprod Dev* **86**:145–155 [Google Scholar](#)
61. Guo S, et al. (2021) **Hypoxia and its possible relationship with endometrial receptivity in adenomyosis: a preliminary study** *Reprod Biol Endocrinol* **19**:7 [Google Scholar](#)
62. Chen M, et al. (2020) **Bioinformatic analysis reveals the importance of epithelial-mesenchymal transition in the development of endometriosis** *Sci Rep* **10**:8442 [Google Scholar](#)
63. Gjorevski N, et al. (2022) **Tissue geometry drives deterministic organoid patterning** *Science* **375**:eaaw9021 [Google Scholar](#)

Author information

Yu Zhang*

State Key Laboratory of Reproductive Medicine and Offspring Health, Center for Reproductive Medicine, Institute of Women, Children and Reproductive Health, Cheeloo College of Medicine, Shandong University, Nanjing, China, National Research Center for Assisted Reproductive Technology and Reproductive Genetics, Shandong University, Jinan, China, Key Laboratory of Reproductive Endocrinology (Shandong University), Ministry of Education, Jinan, China, Shandong Technology Innovation Center for Reproductive Health, Jinan, China, Shandong Provincial Clinical Research Center for Reproductive Health, Jinan, China

*These authors contributed equally

Rusong Zhao*

State Key Laboratory of Reproductive Medicine and Offspring Health, Center for Reproductive Medicine, Institute of Women, Children and Reproductive Health, Cheeloo College of Medicine, Shandong University, Nanjing, China, The Affiliated Suzhou Hospital of Nanjing Medical University, Suzhou Municipal Hospital, Gusu School, Nanjing Medical University, Suzhou, China, National Research Center for Assisted Reproductive Technology and Reproductive Genetics, Shandong University, Jinan, China, Key Laboratory of Reproductive Endocrinology (Shandong University), Ministry of Education, Jinan, China, Shandong Technology Innovation Center for Reproductive Health, Jinan, China, Shandong Provincial Clinical Research Center for Reproductive Health, Jinan, China, Shandong Key Laboratory of Reproductive Medicine, Shandong Provincial Hospital Affiliated to Shandong First Medical University, Jinan, China, Research Unit of Gametogenesis and Health of ART-Offspring, Chinese Academy of Medical Sciences (No.2021RU001), Jinan, China

*These authors contributed equally

Chaoyan Yang

State Key Laboratory of Reproductive Medicine and Offspring Health, Center for Reproductive Medicine, Institute of Women, Children and Reproductive Health, Cheeloo

College of Medicine, Shandong University, Nanjing, China, National Research Center for Assisted Reproductive Technology and Reproductive Genetics, Shandong University, Jinan, China, Key Laboratory of Reproductive Endocrinology (Shandong University), Ministry of Education, Jinan, China, Shandong Technology Innovation Center for Reproductive Health, Jinan, China, Shandong Provincial Clinical Research Center for Reproductive Health, Jinan, China

Jinzhu Song

State Key Laboratory of Reproductive Medicine and Offspring Health, Center for Reproductive Medicine, Institute of Women, Children and Reproductive Health, Cheeloo College of Medicine, Shandong University, Nanjing, China, National Research Center for Assisted Reproductive Technology and Reproductive Genetics, Shandong University, Jinan, China, Key Laboratory of Reproductive Endocrinology (Shandong University), Ministry of Education, Jinan, China, Shandong Technology Innovation Center for Reproductive Health, Jinan, China, Shandong Provincial Clinical Research Center for Reproductive Health, Jinan, China

Peishu Liu

Department of Obstetrics and Gynecology, Qilu Hospital of Shandong University, Jinan, China

Yan Li

State Key Laboratory of Reproductive Medicine and Offspring Health, Center for Reproductive Medicine, Institute of Women, Children and Reproductive Health, Cheeloo College of Medicine, Shandong University, Nanjing, China, National Research Center for Assisted Reproductive Technology and Reproductive Genetics, Shandong University, Jinan, China, Key Laboratory of Reproductive Endocrinology (Shandong University), Ministry of Education, Jinan, China, Shandong Technology Innovation Center for Reproductive Health, Jinan, China, Shandong Provincial Clinical Research Center for Reproductive Health, Jinan, China, Shandong Key Laboratory of Reproductive Medicine, Shandong Provincial Hospital Affiliated to Shandong First Medical University, Jinan, China, Research Unit of Gametogenesis and Health of ART-Offspring, Chinese Academy of Medical Sciences (No.2021RU001), Jinan, China

Boyang Liu

State Key Laboratory of Reproductive Medicine and Offspring Health, Center for Reproductive Medicine, Institute of Women, Children and Reproductive Health, Cheeloo College of Medicine, Shandong University, Nanjing, China, National Research Center for Assisted Reproductive Technology and Reproductive Genetics, Shandong University, Jinan, China, Key Laboratory of Reproductive Endocrinology (Shandong University), Ministry of Education, Jinan, China, Shandong Technology Innovation Center for Reproductive Health, Jinan, China, Shandong Provincial Clinical Research Center for Reproductive Health, Jinan, China, Shandong Key Laboratory of Reproductive Medicine, Shandong Provincial Hospital Affiliated to Shandong First Medical University, Jinan, China, Research Unit of Gametogenesis and Health of ART-Offspring, Chinese Academy of Medical Sciences (No.2021RU001), Jinan, China

ORCID iD: [0000-0002-1285-4842](https://orcid.org/0000-0002-1285-4842)

Tao Li

Department of Gynecology, Shandong Provincial Hospital, Cheeloo College of Medicine, Shandong University, Jinan, China, Department of Gynecology, Shandong Provincial Hospital Affiliated to Shandong First Medical University, Jinan, China

Changjian Yin

State Key Laboratory of Reproductive Medicine and Offspring Health, Center for Reproductive Medicine, Institute of Women, Children and Reproductive Health, Cheeloo College of Medicine, Shandong University, Nanjing, China, National Research Center for Assisted Reproductive Technology and Reproductive Genetics, Shandong University, Jinan, China, Key Laboratory of Reproductive Endocrinology (Shandong University), Ministry of Education, Jinan, China, Shandong Technology Innovation Center for Reproductive Health, Jinan, China, Shandong Provincial Clinical Research Center for Reproductive Health, Jinan, China

Minghui Lu

State Key Laboratory of Reproductive Medicine and Offspring Health, Center for Reproductive Medicine, Institute of Women, Children and Reproductive Health, Cheeloo College of Medicine, Shandong University, Nanjing, China, National Research Center for Assisted Reproductive Technology and Reproductive Genetics, Shandong University, Jinan, China, Key Laboratory of Reproductive Endocrinology (Shandong University), Ministry of Education, Jinan, China, Shandong Technology Innovation Center for Reproductive Health, Jinan, China, Shandong Provincial Clinical Research Center for Reproductive Health, Jinan, China

Zhenzhen Hou

State Key Laboratory of Reproductive Medicine and Offspring Health, Center for Reproductive Medicine, Institute of Women, Children and Reproductive Health, Cheeloo College of Medicine, Shandong University, Nanjing, China, National Research Center for Assisted Reproductive Technology and Reproductive Genetics, Shandong University, Jinan, China, Key Laboratory of Reproductive Endocrinology (Shandong University), Ministry of Education, Jinan, China, Shandong Technology Innovation Center for Reproductive Health, Jinan, China, Shandong Provincial Clinical Research Center for Reproductive Health, Jinan, China

Chuanxin Zhang

State Key Laboratory of Reproductive Medicine and Offspring Health, Center for Reproductive Medicine, Institute of Women, Children and Reproductive Health, Cheeloo College of Medicine, Shandong University, Nanjing, China, National Research Center for Assisted Reproductive Technology and Reproductive Genetics, Shandong University, Jinan, China, Key Laboratory of Reproductive Endocrinology (Shandong University), Ministry of Education, Jinan, China, Shandong Technology Innovation Center for Reproductive Health, Jinan, China, Shandong Provincial Clinical Research Center for Reproductive Health, Jinan, China

Zi-Jiang Chen

State Key Laboratory of Reproductive Medicine and Offspring Health, Center for Reproductive Medicine, Institute of Women, Children and Reproductive Health, Cheeloo College of Medicine, Shandong University, Nanjing, China, National Research Center for Assisted Reproductive Technology and Reproductive Genetics, Shandong University, Jinan, China, Key Laboratory of Reproductive Endocrinology (Shandong University), Ministry of Education, Jinan, China, Shandong Technology Innovation Center for Reproductive Health, Jinan, China, Shandong Provincial Clinical Research Center for Reproductive Health, Jinan, China, Shandong Key Laboratory of Reproductive Medicine, Shandong Provincial Hospital Affiliated to Shandong First Medical University, Jinan, China, Research Unit of Gametogenesis and Health of ART-Offspring, Chinese Academy of Medical Sciences (No.2021RU001), Jinan, China, Shanghai Key Laboratory for Assisted Reproduction and Reproductive Genetics,

Shanghai, China, Department of Reproductive Medicine, Ren Ji Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai, China

For correspondence: chenzijiang@hotmail.com

Keliang Wu

State Key Laboratory of Reproductive Medicine and Offspring Health, Center for Reproductive Medicine, Institute of Women, Children and Reproductive Health, Cheeloo College of Medicine, Shandong University, Nanjing, China, National Research Center for Assisted Reproductive Technology and Reproductive Genetics, Shandong University, Jinan, China, Key Laboratory of Reproductive Endocrinology (Shandong University), Ministry of Education, Jinan, China, Shandong Technology Innovation Center for Reproductive Health, Jinan, China, Shandong Provincial Clinical Research Center for Reproductive Health, Jinan, China, Shandong Key Laboratory of Reproductive Medicine, Shandong Provincial Hospital Affiliated to Shandong First Medical University, Jinan, China, Research Unit of Gametogenesis and Health of ART-Offspring, Chinese Academy of Medical Sciences (No.2021RU001), Jinan, China

For correspondence: wukeliang_527@163.com

Han Zhao

State Key Laboratory of Reproductive Medicine and Offspring Health, Center for Reproductive Medicine, Institute of Women, Children and Reproductive Health, Cheeloo College of Medicine, Shandong University, Nanjing, China, National Research Center for Assisted Reproductive Technology and Reproductive Genetics, Shandong University, Jinan, China, Key Laboratory of Reproductive Endocrinology (Shandong University), Ministry of Education, Jinan, China, Shandong Technology Innovation Center for Reproductive Health, Jinan, China, Shandong Provincial Clinical Research Center for Reproductive Health, Jinan, China, Shandong Key Laboratory of Reproductive Medicine, Shandong Provincial Hospital Affiliated to Shandong First Medical University, Jinan, China, Research Unit of Gametogenesis and Health of ART-Offspring, Chinese Academy of Medical Sciences (No.2021RU001), Jinan, China

For correspondence: hanzh80@sdu.edu.cn

Editors

Reviewing Editor

Wei Yan

Washington State University, Pullman, United States of America

Senior Editor

Wei Yan

Washington State University, Pullman, United States of America

Reviewer #1 (Public review):

Summary:

This study generated 3D cell constructs from endometrial cell mixtures that were seeded in the Matrigel scaffold. The cell assemblies were treated with hormones to induce a "window of implantation" (WOI) state. Although many bioinformatic analyses point in this direction, there are major concerns that must be addressed.

Strengths:

The addition of 3 hormones to enhance the WOI state (although not clearly supported in comparison to the secretory state).

Comments on revisions:

The authors did their best to revise their study according to the Reviewers' comments. However, the study remains unconvincing, incomplete and at the same time still too dense and not focused enough.

<https://doi.org/10.7554/eLife.90729.3.sa2>

Reviewer #2 (Public review):

Zhang et al. have developed an advanced three-dimensional culture system of human endometrial cells, termed a receptive endometrial assembloid, that models the uterine lining during the crucial window of implantation (WOI). During this mid-secretory phase of the menstrual cycle, the endometrium becomes receptive to an embryo, undergoing distinctive changes. In this work, endometrial cells (epithelial glands, stromal cells, and immune cells from patient samples) were grown into spheroid assembloids and treated with a sequence of hormones to mimic the natural cycle. Notably, the authors added pregnancy-related factors (such as hCG and placental lactogen) on top of estrogen and progesterone, pushing the tissue construct into a highly differentiated, receptive state. The resulting WOI assembloid closely resembles a natural receptive endometrium in both structure and function. The cultures form characteristic surface structures like pinopodes and exhibit abundant motile cilia on the epithelial cells, both known hallmarks of the mid-secretory phase. The assembloids also show signs of stromal cell decidualization and an epithelial mesenchymal transition, like process at the implantation interface, reflecting how real endometrial cells prepare for possible embryo invasion.

Although the WOI assembloid represents an important step forward, it still has limitations: the supportive stromal and immune cell populations decrease over time in culture, so only early-passage assembloids retain full complexity. Additionally, the differences between the WOI assembloid and a conventional secretory-phase organoid are more quantitative than absolute; both respond to hormones and develop secretory features, but the WOI assembloid achieves a higher degree of differentiation due to the addition of "pregnancy" signals. Overall, while it's a reinforced model (not an exact replica of the natural endometrium), it provides a valuable in vitro system for implantation studies and testing potential interventions, with opportunities to improve its long-term stability and biological fidelity in the future.

<https://doi.org/10.7554/eLife.90729.3.sa1>

Author response:

The following is the authors' response to the previous reviews

We have thoroughly addressed all the reviewers' comments and meticulously revised the manuscript. Key modifications include the following:

(a) Organizing the Logic and Highlighting Key Findings: We have revised the manuscript to emphasize key findings (especially the distinctions between the SEC and WOI groups) according to the following logic: constructing a receptive endometrial organoid, comparing

its molecular characteristics with those of the receptive endometrium, highlighting its main features (hormone response, enhanced energy metabolism, ciliary assembly and motility, epithelial-mesenchymal transition), and exploring the function involved in embryo interaction.

(b) Clarity and Better Description of Bioinformatic Analyses: We have revised the sections involving bioinformatic analyses to provide a more streamlined and comprehensible explanation. Instead of overwhelming the reader with excessive details, we focused on the most important findings, and performed additional experimental validation.

(c) Rationale for Gene Selection: We have clarified the rationale for selecting certain genes and pathways for inclusion in the analysis and manuscript. The associated gene expression data for all figures have been provided in the attached Dataset.

(d) In the response letter, we have provided the detailed presentation of the methodological optimization for constructing this endometrial assembloids, along with optimization and comparison of endometrial organoid culture media. Furthermore, in the Limitations section, we have explicitly stated that stromal cells and immune cells gradually diminish with increasing passage numbers. Therefore, this study primarily utilized endometrial assembloids within the first three passages for all investigations.

Below, we provide a point-by-point response to each comment, with all modifications highlighted in the revised manuscript. We respectfully hope that these revisions effectively address the concerns raised by the reviewers.

Public Reviews:

Reviewer #1 (Public Review):

This study generated 3D cell constructs from endometrial cell mixtures that were seeded in the Matrigel scaffold. The cell assemblies were treated with hormones to induce a "window of implantation" (WOI) state. The authors did their best to revise their study according to the reviewers' comments. However, the study remains unconvincing and at the same time too dense and not focused enough.

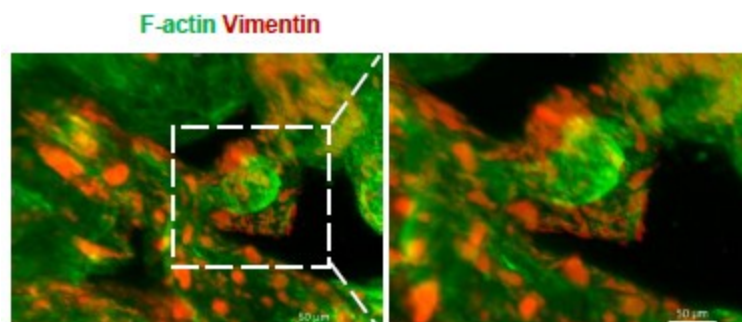
(1) The use of the term organoids is still confusing and should be avoided. Organoids are epithelial tissue-resembling structures. Hence, the multicell aggregates developed here are rather "coculture models" (or "assembloids"). It is still unexpected (unlikely) that these structures containing epithelial, stromal and immune cells can be robustly passaged in the epithelial growth conditions used. All other research groups developing real organoids from endometrium have shown that only the epithelial compartment remains in culture at passaging (while the stromal compartment is lost). If authors keep to their idea, they should perform scRNA-seq on both early and late (passage 6-10) "organoids". And they should provide details of culturing/passaging/plating etc that are different with other groups and might explain why they keep stromal and immune cells in their culture for such a long time. In other words, they should then in detail compare their method to the standard method of all other researchers in the field, and show the differences in survival and growth of the stromal and immune cells.

(1) We appreciate your feedback and have revised the term 'organoids' to 'assembloids'. 2)

I. Due to budget constraints, this study did not perform scRNA-seq on both early and late passages (P6-P10). Instead, immunofluorescence staining confirmed the persistence of stromal cells at passage 6 (as shown below).

Author response image 1.

Whole-mount immunofluorescence showed that Vimentin+ F-actin+ cells (stromal cells) were arranged around the glandular spheres that were only F-actin+(passage 6).



II. Improvements in this study include the following.

- a. Optimization of endometrial tissue processing: The procedures for tissue collection, pretreatment, digestion, and culture were refined to maximize the retention of endometrial epithelial cells, stromal cells, and immune cells (detailed optimizations are provided in Response Table 1).
- b. Enhanced culture medium formulation: Based on previous protocols, WNT3A was added to promote organoid development and differentiation (PMID: 27315476), while FGF2 was supplemented to improve stromal cell survival (PMID: 35224622) (see Response Table 2 for medium comparisons). Representative culture outcomes are shown in the figure below.

We acknowledge that the stromal and immune cells in this system still exhibit differences compared to their *in vivo* counterparts. In this study, we utilized the first three passages, which offer optimal cell diversity and viability, to meet experimental needs. However, replicating and maintaining the full complexity of endometrial cell types *in vitro* remains a major challenge in the field—one that we are actively working to address.

Author response table 1.

Methodological Optimization of Endometrial Organoids (Construction, Passaging, and Cryopreservation)

Protocol	Key Steps	2017 Nat Cell Biol	2017 Development	2019 PNAS	2021 eLife	This Research
Construction	Remove blood and debris	room temperature	-	37 °C	-	4°C
	Enzyme	collagenase V, dispace II	collagenase IV, TrypLE	collagenase V, dispace II	collagenase I, DNase I	collagenase V, dispace II, DNase I
	Digestion	NA	1.5-2.5h	40-50min	1h	20min
	Stand	2min	-	2min	-	-
	Cell strainers	100µm	70 µm	100µm	40 µm	40 µm
	Matrigel	20 x volume (to pellet)	70%	20 x volume (to pellet)	20 x volume (to pellet)	75%
	Centrifugation	500g, 5 min	190g, 5 min, 4°C	-	600g 5min	400g, 5 min
	Assembloids	-	-	-	EnSC+ gland	-
Passaging	Organoid harvesting	scrape	TrypLE	scrape	scrape	collagenase V, dispace II, DNase I
Cryopreservation	Freezing medium	10% Serum, 10% DMSO	-	90% FBS, 10% DMSO	-	no serum, 10%DMSO

Author response table 2.

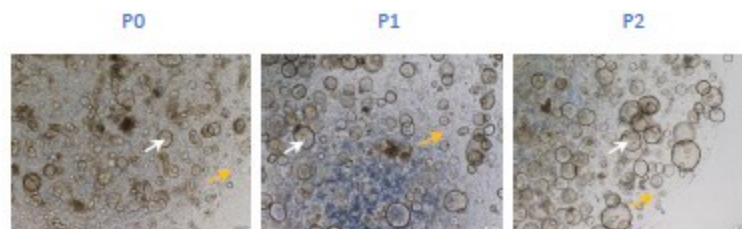
Optimization and comparison of endometrial organoid culture media

Component	2017 Nat Cell Biol	2017 Development	2021 eLife	This Research
Medium	Advanced DMEM/F12	DMEM/F12	Advanced DMEM/F12	DMEM/F12 (Phenol red-free)
N2	1%	1%	1%	1%
B27	2%	2%	2%	2%
Primocin	100µg/ml	1%	1%	1%
N-Acetyl-L-cysteine	1.25 mM	1.25 mM	1.25 mM	1.25 mM

Component	2017 Nat Cell Biol	2017 Development	2021 eLife	This Research
L-glutamine	2mM	2 mM	2 mM	GlutaMAX™ (100 X)
EGF	50 ng/ml	50 ng/ml	50 ng/ml	50 ng/ml
Noggin	100 ng/ml	100 ng/ml	100 ng/ml	100 ng/ml
Rspondin-1	500ng/ml	200 ng/ml (some experiments)	500 ng/ml	200 ng/ml
FGF-10	100ng/ml	50ng/ml	100 ng/ml	-
HGF	50ng/ml	-	50 ng/ml	-
A83-01	500nM	500nM	500 nM	500nM
Nicotinamide	10nM	1 mM	10 mM	1 mM
ITS	-	1%	ITS-X	1%
WNT-3A	-	200 ng/ml (some experiments)	-	200 ng/ml
p38 inhibitor SB202190	-	10µM	-	10µM
FGF2	-	-	-	100 ng/ml
E2	-	2 nM	1µM	-
MPA	-	-	1µM	-
8-bromo-cAMP	-	-	0.5 mM	-
Non-essential amino acids	-	-	1x	-
Essential amino acids	-	-	1x	-

Author response image 2.

Bright-field microscopy captures the expansion of glands and surrounding stromal cells across passages 0 to 2 (scar bar=200µm) (Yellow arrows: stromal cells; White arrows: glands).



(2) The paper is still much too dense, touching upon all kind of conclusions from the manifold bioinformatic analyses. The latter should be much clearer and better described, and then some interesting findings (pathways/genes) should be highlighted without mentioning every single aspect that is observed. The paper needs a lot of editing to better focus and extract take-home messages, not bombing the reader with a mass of pathways, genes etc which makes the manuscript just not readable or 'digest-able'. There is no explanation whatever and no clear rationale why certain genes are included in a list while others are not. There is the impression that mass bioinformatics is applied without enough focus.

Thanks for your suggestions. We have made improvements and revisions in the following areas:

(a) Clarity and Better Description of Bioinformatic Analyses: We have revised the sections involving bioinformatic analyses to provide a more streamlined and comprehensible explanation. Instead of overwhelming the reader with excessive details, we focused on the most important findings.

(b) Organizing the Logic and Highlighting Key Findings: We have revised the manuscript to emphasize key findings according to the following logic: constructing a receptive endometrial organoid, comparing its molecular characteristics with those of the receptive endometrium, highlighting its main features (hormone response, enhanced energy metabolism, ciliary assembly and motility, epithelial-mesenchymal transition), and exploring the function involved in embryo interaction.

(c) Rationale for Gene Selection: We have clarified the rationale for selecting certain genes and pathways for inclusion in the analysis and manuscript.

We hope these revisions address your concerns and improve the overall quality and clarity of the manuscript. Thank you once again for your valuable input.

(3) The study is much too descriptive and does not show functional validation or exploration (except glycogen production). Some interesting findings extracted from the bioinformatics must be functionally tested.

Thanks for your suggestions. We have restructured the logic and revised the manuscript, incorporating functional validation. The focus is on the following points: highlighting its main features (hormone response, enhanced energy metabolism, ciliary assembly and motility, epithelial-mesenchymal transition), and exploring the functions involved in embryo interaction.

(4) In contrast to what was found in vivo (Wang et al. 2020), no abrupt change in gene expression pattern is mentioned here from the (early-) secretory to the WoI phase. Should be discussed. Although the bioinformatic analyses point into this direction, there are major concerns which must be solved before the study can provide the needed reliability and credibility for revision.

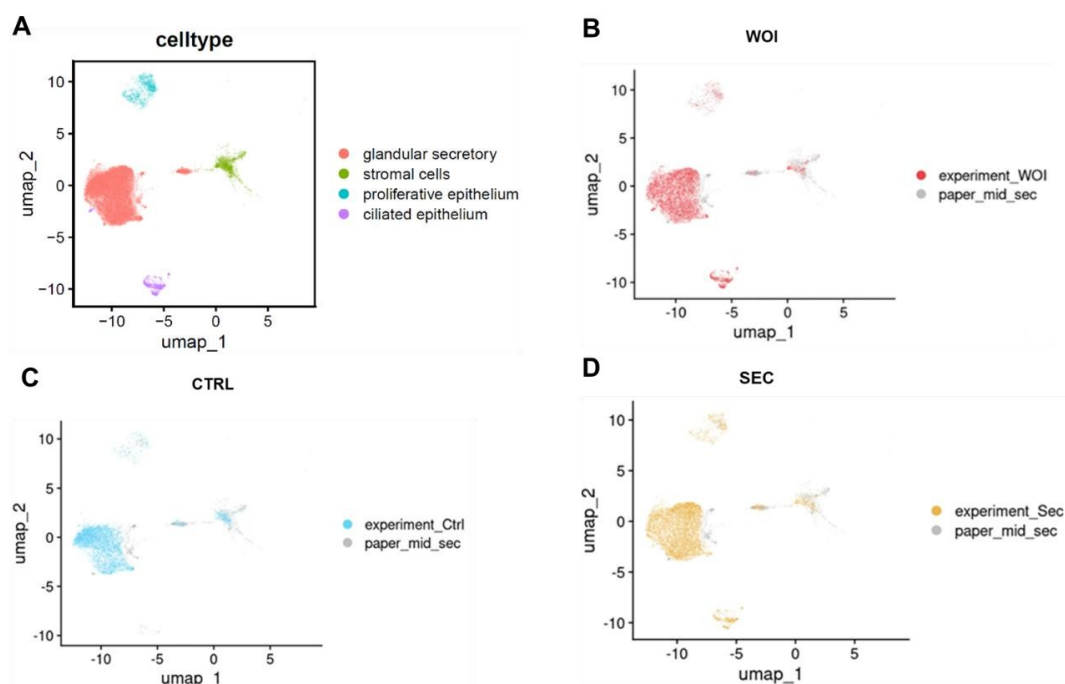
To further investigate the abrupt change, the Mfuzz algorithm was utilized to analyze gene expression across the three groups, focusing on gene clusters that were progressively upregulated or downregulated. It was observed that mitochondrial and cilia-related genes exhibited the highest expression levels in WOI endometrial organoids, as well as cell junction and negative regulation of cell differentiation were downregulated (Figure 4A).

(5) All data should be benchmarked to the Wang et al 2020 and Garcia-Alonso et al. 2021 papers reporting very detailed scRNA-seq data, and not only the Stephen R. Quake 2020 paper.

We appreciate your suggestion. By integrating data from Garcia-Alonso et al. (2021) (shown in the figure below), we observed that both WOI organoids and SEC organoids exhibit increased glandular secretory epithelium and developed ciliated epithelium, mirroring features of mid-secretory endometrium. The findings exhibit parallels when contrasting these two papers.

Author response image 3.

UMAP visualization of integrated scRNA-seq data (our dataset and Garcia-Alonso et al. 2021) showing: (A) cell types, (B) WOI-org, (C) CTRL-org, (D) SEC-org versus published midsecretory samples.



(6) Fig. 2B: Vimentin staining is not at all clear. F-actin could be used to show the typical morphology of the stromal cells?

We appreciate your suggestion. We performed additional staining for F-actin based on Vimentin, and found that Vimentin+ F-actin+ cells (stromal cells) were arranged around the glandular spheres that were only F-actin+.

(7) Where does the term "EMT-derived stromal cells" come from? On what basis has this term been coined?

Within endometrial biology, stromal cells in the transition from epithelial to mesenchymal phenotype are specifically referred to as 'stromal EMT transition cells' (PMID: 39775038, PMID: 39968688).

In certain cancers or fibrotic diseases, epithelial cells can transition into a mesenchymal phenotype, contributing to the stromal environment that supports tumor growth or tissue remodeling (PMID: 20572012).

| (8) CD44 is shown in Fig. 2D but the text mentions CD45 (line 159)?

In Fig 2D, T cells are defined as a cluster of CD45+CD3+ cells, further subdivided into CD4+ and CD8+ T cells based on their expression of CD4 and CD8. This figure does not include data on CD44.

| (9) All quantification experiments (of stainings etc) should be in detail described how this was done. It looks very difficult (almost not feasible) when looking at the provided pictures to count the stained cells.

a. Manual Measurement:

For TEM-observed pinopodes, glycogen particles, microvilli, and cilia, manual region-of-interest (ROI) selection was performed using ImageJ software for quantitative analysis of counts, area, and length. Twenty randomly selected images per experimental group were analyzed for each morphological parameter.

b. Automated Measurement:

We quantified the fluorescence images using ImageJ. Firstly, preprocess them by adjusting brightness and contrast, and removing background noise with the “Subtract Background” feature.

Secondly, set the threshold to highlight the cells, then select the regions of interest (ROI) using selection tools. Thirdly, as for counting the cells, navigate to Analyze > Analyze Particles. AS for measuring the influence intensity and area, set the “Measurement” options as mean gray value. Adjust parameters as needed, and view results in the “Results” window. Save the data for further analysis and ensure consistency throughout your measurements for reliable results.

For 3D fluorescence quantification, ZEN software (Carl Zeiss) was exclusively used, with 11 images analyzed per experimental group. This part has been incorporated into “Supporting Information”

Line 94-100.

c. Normalization Method :

For fluorescence quantification, DAPI was used as an internal reference for normalization, where both DAPI and target fluorescence channel intensities were quantified simultaneously. The normalized target signal intensity (target/DAPI ratio) was then compared across experimental groups. A minimum of 15 images were analyzed for each parameter per group. This part has been incorporated into “Supporting Information” Line 101-104.

| (10) Fig. 3C: it is unclear how quantification can be reliably done. Moreover, OLFM4 looks positive in all cells of Ctrl, but authors still see an increase?

(a) Fluorescence images were quantitatively analyzed using ImageJ by measuring the mean gray values. For normalization, DAPI staining served as an internal reference, with simultaneous measurement of mean gray values in both the target fluorescence channel and the DAPI channel. The relative fluorescence intensity was then calculated as the ratio of target channel to DAPI signal for inter-group quantitative comparisons.

(b) OLFM4 is an E2-responsive gene. Its expression in endometrial organoids of the CTRL group is physiologically normal (PMID: 31666317). However, its fluorescence intensity (quantified as mean gray value) was significantly stronger in both the SEC and WOI groups compared to the CTRL group (quantitative method as described above).

(11) Fig. 3F: Met is downregulated which is not in accordance with the mentioned activation of the PI3K-AKT pathway.

We appreciate your careful review. Our initial description was imprecise. In the revised manuscript, this statement has been removed entirely.

(12) Lines 222-226: transcriptome and proteome differences are not significant; so, how meaningful are the results then? Then, it is very hard to conclude an evolution from secretory phase to WoI.

We appreciate your feedback. The manuscript has been comprehensively revised, and the aforementioned content has been removed.

(13) WoI organoids show an increased number of cilia. However, some literature shows the opposite, i.e. less ciliated cells in the endometrial lining at WoI (to keep the embryo in place). How to reconcile?

Thank you for raising this question. We conducted a statistical analysis of the proportion of ciliated cells across endometrial phases.

(a) Based on the 2020 study by Stephen R. Quake and Carlos Simon's team published in Nature Medicine (PMID: 32929266), the mid-secretory phase (Days 19–23) exhibited a higher proportion of ciliated cells compared to the early-secretory (Days 15–18) and late-secretory phases (Days 24–28) (Fig. R13 A).

(b) According to the 2021 study by Roser Vento-Tormo's team in Nature Genetics, ciliated cell abundance peaked in the early-to-mid-secretory endometrium across all phases (Fig. R13 B-C).

Data were sourced from the Reproductive Cell Atlas.

(14) How are pinopodes distinguished from microvilli? Moreover, Fig. 3 does not show the typical EM structure of cilia.

Thank you for this insightful question.

(a) Pinopodes are large, bulbous protrusions with a smooth apical membrane. Under transmission electron microscopy (TEM), it can be observed that the pinopodes contain various small particles, which are typically extracellular fluid and dissolved substances.

Microvilli are elongated, finger-like projections that typically exhibit a uniform and orderly arrangement, forming a "brush border" structure. Under transmission electron microscopy, dense components of the cytoskeleton, such as microfilaments and microtubules, can be seen at the base of the microvilli.

(b) You may refer to the ciliated TEM structure shown in the current manuscript's Fig. 2E (originally labeled as Fig. 2H in the draft). The cilium is composed of microtubules. The cross-section shows that the periphery of the cilium is surrounded by nine pairs of microtubules arranged in a ring. The longitudinal section shows that the cilium has a long cylindrical structure, with the two central microtubules being quite prominent and located at the center of the cilium.

(15) *There is a recently published paper demonstrating another model for implantation. This paper should be referenced as well (Shibata et al. Science Advances, 2024).*

Thanks for your valuable comments. We have cited this reference in the manuscript at Line 77-78.

(16) *Line 78: two groups were the first here (Turco and Borreto) and should both be mentioned.*

Thanks for your valuable comments. We have cited this reference in the manuscript at Line 74-76.

(17) *Line 554: "as an alternative platform" - alternative to what? Authors answer reviewers' comments by just changing one word, but this makes the text odd.*

Thank you for your review. Here, we propose that this WOI organoid serves as an alternative research platform for studying endometrial receptivity and maternal-fetal interactions, compared to current secretory-phase organoids. In the revised manuscript, we have supplemented the data by co-culturing this WOI organoid with blastoid, demonstrating its robust embryo implantation potential.

Reviewer #2 (Public Review):

In this research, Zhang et al. have pioneered the creation of an advanced organoid culture designed to emulate the intricate characteristics of endometrial tissue during the crucial Window of Implantation (WOI) phase. Their method involves the incorporation of three distinct hormones into the organoid culture, coupled with additives that replicate the dynamics of the menstrual cycle. Through a series of assays, they underscore the striking parallels between the endometrial tissue present during the WOI and their crafted organoids. Through a comparative analysis involving historical endometrial tissue data and control organoids, they establish a system that exhibits a capacity to simulate the intricate nuances of the WOI. The authors made a commendable effort to address the majority of the statements. Developing an endometrial organoid culture methodology that mimics the window of implantation is a game-changer for studying the implantation process. However, the authors should strive to enhance the results to demonstrate how different WOI organoids are from SEC organoids, ensuring whether they are worth using in implantation studies, or a proper demonstration using implantation experiments.

Thank you for your valuable suggestions. The WOI organoids differ from SEC organoids in the following aspects.

(1) Structurally, WOI endometrial organoids exhibit subcellular features characteristic of the implantation window: densely packed pinopodes on the luminal side of epithelial cells, abundant glycogen granules, elongated and tightly arranged microvilli, and increased cilia (Figure 2F).

(2) At the molecular level, WOI organoids show enlarged and functionally active mitochondria, enhanced ciliary assembly and motility, and single-cell signatures resembling mid-secretory endometrium.

Specifically, mitochondrial- and cilia-related genes/proteins are most highly expressed in WOI organoids (Figure 4A,B). TEM analysis revealed that WOI organoids have the largest average mitochondrial area (Figure 4C). Mitochondrial-related genes display an increasing trend across the three organoid groups, and WOI organoids produce more ATP and IL-8 (Figure 4D,E).

For cilia, WOI organoids upregulate genes/proteins involved in ciliary assembly, basal bodies, and motile cilia, while downregulating non-motile cilia markers (Figure 5A-C).

Single-cell analysis further confirms that WOI organoids recapitulate mid-secretory endometrial traits in mitochondrial metabolism and cell adhesion (Figure 2G).

(3) Functionally, WOI organoids demonstrate superior embryo implantation potential. Given the scarcity and ethical constraints of human embryos, we used blastoids for implantation assays (Figure 6A). These blastoids successfully grew within endometrial organoids, established interactions (Figure 6B), and exhibited normal trilineage differentiation (epiblast: OCT4; hypoblast: GATA6; trophoblast: KRT18) (Figure 6C). WOI organoids achieved significantly higher blastoid survival (66% vs. 19% in CTRL and 28% in SEC) and interaction rates (90% vs. 47% in CTRL and 53% in SEC), confirming their robust embryo-receptive capacity (Figure 6D,E).

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

In conclusion, it is needed to first meet all the concerns of the reviewers and then submit an appropriately adapted and comprehensive paper (also showing the robustness of the "organoids" and functionality of the findings) instead of this still fully descriptive paper. Further comments are included in the rebuttal document of the authors and will be provided by the editor as PDF.

Reviewer #2 (Recommendations For The Authors):

The authors made a good effort to reply all the statements. However, there are some points that the authors need to address.

- *There is an inconsistency in the manuscript regarding the number of passages in which the organoids are used; in the response to the reviewers, it mentions 5 passages, while in the Materials and Methods section, it states 3 passages.*

We sincerely appreciate your thorough review. In this study, organoids within the first three passages were used. To address the reviewer's question comprehensively, we have now provided a detailed account of the organoid passage history in our response.

- *We agree that the difference between SEC and WOI organoids may be subtle, but in response to this, the authors should explain what they mean by "the most notable differences lie in the more comprehensive differentiation and varied cellular functions exhibited by WOI organoids..."*

In the original manuscript, this statement indicated that, at the single-cell level, WOI endometrial organoids exhibited more functionally mature and thoroughly differentiated characteristics compared to SEC endometrial organoids (See details below).

In the revised version, we have restructured this section to focus on following aspects: hormone response, energy metabolism, ciliary assembly and motility, epithelial-mesenchymal transition, and embryo implantation potential. Consequently, the "the most notable differences lie in the more comprehensive differentiation and varied cellular functions exhibited by WOI organoids..." has been removed.

(1) Varied cellular functions:

- a. Secretory Epithelium: Compared to SEC organoids, WOI organoids exhibit enhanced peptide metabolism and mitochondrial energy metabolism in their secretory epithelium, supporting endometrial decidualization and embryo implantation (Figure 3F).
- b. Proliferative Epithelium: Compared to SEC organoids, WOI organoids demonstrate enhanced GTPase activity, angiogenesis, cytoskeletal assembly, cell differentiation, and RAS protein signaling in their proliferative epithelium (Figure S2G).
- c. Ciliated Epithelium: The ciliated epithelium of WOI endometrial organoids is associated with the regulation of vascular development and exhibits higher transcriptional activity compared to SEC organoids (Figure 5E).
- d. Stromal Cells: Compared to SEC organoids, WOI organoids exhibit enhanced cell junctions, cell migration, and cytoskeletal regulation in EMT-derived stromal cells (Figure S4A right panel). Similarly, cell junctions are also strengthened in stromal cells (Figure S4A left panel).

(2) comprehensive differentiation:

- a. Compared to SEC organoids, WOI organoids exhibit more complete differentiation from proliferative epithelium to secretory epithelium (Figure 3G).
- b. The WOI organoids demonstrate more robust ciliary differentiation compared to SEC organoids (Figure 5F).
- c. The proliferative epithelium progressively differentiates into EMT-derived cells. Compared to SEC organoids, WOI organoids are predominantly localized at the terminal end of the differentiation trajectory, indicating more complete differentiation (Figure S4B).

• What do the authors mean by "average intensity" when referring to the extra reagents added to the WOI? The results that the authors show in response to Reviewer 2's Q1 must be included as part of the results and explain how it was done in the materials and methods section.

This parameter indicates the growth status of organoids. It measures the gray value of organoids through long-term live-cell tracking. When organoids undergo apoptosis, they progressively condense into denser solid spheres, leading to an increase in gray value (average intensity). This content has been incorporated into the Results section (Line 129) and is further explained in the Supporting Information "Materials and Methods" (Lines 70-77).

• In panel 1C, it is not possible to see the stromal cells around because they are brightfield images.

You are partly right. Bright-field images alone indeed make it difficult to distinguish stromal cells. However, by combining whole-mount immunofluorescence staining with the characteristic elongated spindle-shaped morphology of stromal cells, we were able to roughly determine their distribution in the bright-field images.

• *Responding to Reviewer 2's question Q7, the authors indicate how they establish the cluster. However, they do not specify whether they extrapolate the data from a database or create the cluster themselves based on the literature. It should be stated from which classification list (or classification database) the extrapolation has been made.*

Within endometrial biology, stromal cells in the transition from epithelial to mesenchymal phenotype are specifically referred to as 'stromal EMT transition cells' (PMID: 39775038, PMID: 39968688).

In certain cancers or fibrotic diseases, epithelial cells can transition into a mesenchymal phenotype, contributing to the stromal environment that supports tumor growth or tissue remodeling (PMID: 20572012).

• *Regarding Reviewer 2's question Q8, if the authors have not been able to make comparisons with, at least, SEC organoids, unfortunately, the ERT loses much of its strength and should not serve as support.*

We agree with you at this point. These results have been moved to the supplementary figures.

• *If the differences in the transcriptome and proteome between SEC and WOI organoids are not significant, the result does not support the authors' model. If there are barely any differences at the proteome and transcriptome level between SEC and WOI organoids, why would anyone choose to use their model over SEC organoids?*

We sincerely appreciate your valuable feedback. In this revised manuscript, we have further integrated transcriptomic and proteomic analyses, revealing that WOI organoids exhibit enlarged and functionally active mitochondria, along with enhanced cilia assembly and motility compared to SEC organoids. Additionally, using a blastoid model, we demonstrated that WOI organoids possess superior embryo implantation potential, significantly outperforming SEC organoids. Our research group aims to develop an embryo co-culture model. Through systematic comparisons of structural, molecular, and co-culture characteristics between SEC and WOI organoids, we ultimately confirmed the superior performance of WOI organoids.

• *SEC and WOI organoids must be different enough to establish a new model, and the authors do not demonstrate that they are.*

Thank you for your valuable feedback. In the revised manuscript, we have emphasized the distinctions between SEC and WOI organoids in terms of structure, molecular characteristics, and functionality (co-culture with blastoid), as detailed below.

(1) Structurally, WOI endometrial organoids exhibit subcellular features characteristic of the implantation window: densely packed pinopodes on the luminal side of epithelial cells, abundant glycogen granules, elongated and tightly arranged microvilli, and increased cilia (Figure 2F).

(2) At the molecular level, WOI organoids show enlarged and functionally active mitochondria, enhanced ciliary assembly and motility, and single-cell signatures resembling mid-secretory endometrium.

Specifically, mitochondrial- and cilia-related genes/proteins are most highly expressed in WOI organoids (Figure 4A,B). TEM analysis revealed that WOI organoids have the largest average mitochondrial area (Figure 4C). Mitochondrial-related genes display an increasing trend

across the three organoid groups, and WOI organoids produce more ATP and IL-8 (Figure 4D,E).

For cilia, WOI organoids upregulate genes/proteins involved in ciliary assembly, basal bodies, and motile cilia, while downregulating non-motile cilia markers (Figure 5A-C).

Single-cell analysis further confirms that WOI organoids recapitulate mid-secretory endometrial traits in mitochondrial metabolism and cell adhesion (Figure 2G).

(3) Functionally, WOI organoids demonstrate superior embryo implantation potential. Given the scarcity and ethical constraints of human embryos, we used blastoids for implantation assays (Figure 6A). These blastoids successfully grew within endometrial organoids, established interactions (Figure 6B), and exhibited normal trilineage differentiation (epiblast: OCT4; hypoblast: GATA6; trophoblast: KRT18) (Figure 6C). WOI organoids achieved significantly higher blastoid survival (66% vs. 19% in CTRL and 28% in SEC) and interaction rates (90% vs. 47% in CTRL and 53% in SEC), confirming their robust embryo-receptive capacity (Figure 6D,E).

• Regarding Q16, Boretto et al. 2017 and Turco et al. 2017 also manage to isolate stromal cells, but they lose them between passages. It's not a matter of isolating them from the tissue or not, but rather how they justify their maintenance in culture. In the images added by the authors, it can be seen that the majority of stromal cells are lost from P0 to P1 after thawing. I still believe that the epithelial part can be passed and maintained, but the rest cannot, and that should be mentioned in the paper as a limitation. However, the authors can demonstrate the maintenance of stromal cells by performing immunostaining with vimentin from passages 4, 5, and 6.

Thank you for your valuable comments. We have added the statement 'Stromal cells and immune cells are difficult to pass down stably and their proportion is lower than that in the in vivo endometrium' to the Limitations section (Line 364-365). Additionally, we performed immunostaining with vimentin starting from passage 6 and confirmed the presence of Vimentin+ F-actin+ stromal cells (as shown in Author response image 1).

<https://doi.org/10.7554/eLife.90729.3.sa0>